

História da Computação

De 1970 a 1979

8. O Caminho para o Computador Pessoal

- A década de 1970 foi para a computação uma década de ramificação: de um lado o desenvolvimento de supercomputadores cada vez mais poderosos e de outro lado o surgimento do computador de uso pessoal.
- Tudo isso foi possibilitado principalmente pela evolução do hardware. A década de 1970 viu a consolidação do circuito integrado, mas mais do que isso, a criação do microprocessador, ou seja, um circuito integrado que sozinho implementa um computador completo.
- Em relação aos computadores pessoais, apesar dos kits disponibilizados na década de 1950 e 1960, com reativamente pouca aceitação fora do circuito dos hobistas, os anos 1970 viram o surgimento dos primeiros microcomputadores populares, como Kenbak, Xerox Alto, Altair, Apple 1 e CompuColor.
- Especialmente o ano de 1977 é considerado fundamental para a história da computação pessoal porque neste ano foi lançada a “Trindade de 77”, ou seja, três computadores que foram grandes sucessos para o público em geral: o Apple II, o Tandy TRS-80 e o Commodore PET.
- Em termos de software, o final da década viu o surgimento do processador de texto WordStar e da planilha VisiCalc, os quais foram os grandes responsáveis por colocar os microcomputadores também nas empresas e não apenas nos lares.
- No Brasil essa década viu o aparecimento da indústria nacional de computadores, com a criação do Pato Feio em 1972, a partir do qual os engenheiros obtiveram experiência para criar a primeira empresa construtora de computadores nacionais em 1974, a COBRA.
- Ainda nessa década fértil de ideias foram criadas várias linguagens de programação tais como Pascal, Smalltalk, C e Prolog. No final dos anos 1970 a computação era uma área em franco desenvolvimento, e começava a estar presente não apenas em grandes corporações e governo, mas nos lares, escolas e pequenas empresas, e ainda muito haveria por vir.

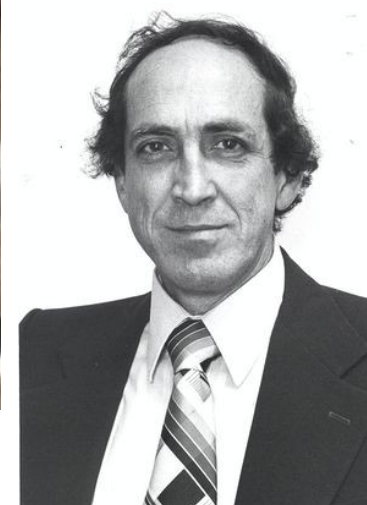
8.1 Memória Intel 1103 – 1970



Esse circuito foi o responsável pelo fim das memórias de núcleo magnético, pois sua produção era muito mais simples. Ao invés de passar fios minúsculos através de pequenos anéis, um circuito 1103 podia ser produzido por um processo semelhante ao da fotografia.



Joel Karp



Robert H. Dennard (Estados Unidos, 1932)



William Regitz

8.2 Pascal – 1970



Nicklaus Wirth (Suíça, 1934)

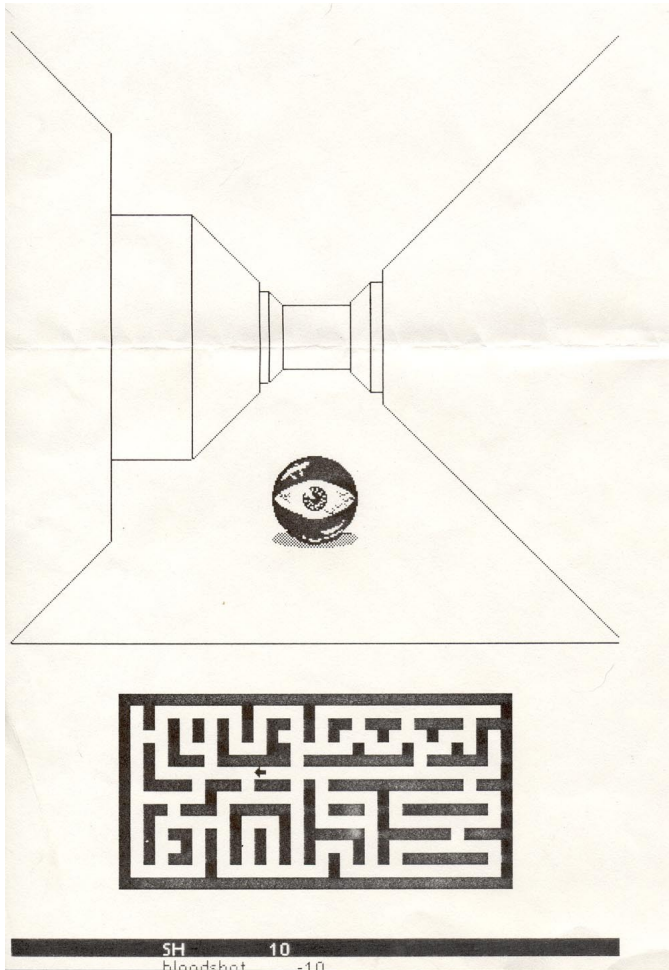
lucid, systematic,
and penetrating
treatment of basic
and dynamic data
structures, sorting,
recursive algorithms,
language structures,
and compiling

NIKLAUS WIRTH

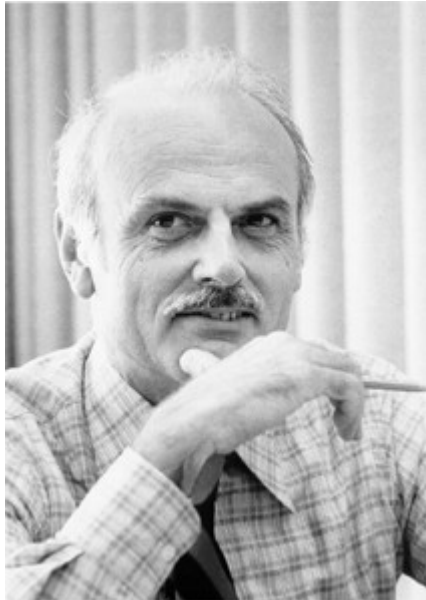
Algorithms + Data Structures = Programs

PRENTICE-HALL
SERIES IN
AUTOMATIC
COMPUTATION

8.3 Imlac PDS-1, o Computador Gráfico – 1970



8.4 Bancos de Dados Relacionais – 1970



Edgar F. Codd (Reino Unido, 1923-2003)

A Relational Model of Data for Large Shared Data Banks

E. F. CODD
IBM Research Laboratory, San Jose, California

Future users of large data banks must be protected from having to know how the data is organized in the machine (the internal representation). A prompting service which supplies such information is not a satisfactory solution. Activities of users at terminals and most application programs should remain unaffected when the internal representation of data is changed and even when some aspects of the external representation are changed. Changes in data representation will often be needed as a result of changes in query, update, and report traffic and natural growth in the types of stored information.

Existing noninferential, formatted data systems provide users with tree-structured files or slightly more general network models of the data. In Section 1, inadequacies of these models are discussed. A model based on n -ary relations, a normal form for data base relations, and the concept of a universal data sublanguage are introduced. In Section 2, certain operations on relations (other than logical inference) are discussed and applied to the problems of redundancy and consistency in the user's model.

KEY WORDS AND PHRASES: data bank, data base, data structure, data organization, hierarchies of data, networks of data, relations, derivability, redundancy, consistency, composition, join, retrieval language, predicate calculus, security, data integrity

CR CATEGORIES: 3.70, 3.73, 3.75, 4.20, 4.22, 4.29

1. Relational Model and Normal Form

1.1. INTRODUCTION

This paper is concerned with the application of elementary relation theory to systems which provide shared access to large banks of formatted data. Except for a paper by Childs [1], the principal application of relations to data systems has been to deductive question-answering systems. Levin and Maron [2] provide numerous references to work in this area.

In contrast, the problems treated here are those of *data independence*—the independence of application programs and terminal activities from growth in data types and changes in data representation—and certain kinds of *data inconsistency* which are expected to become troublesome even in nondeductive systems.

The relational view (or model) of data described in Section 1 appears to be superior in several respects to the graph or network model [3, 4] presently in vogue for non-inferential systems. It provides a means of describing data with its natural structure only—that is, without superimposing any additional structure for machine representation purposes. Accordingly, it provides a basis for a high level data language which will yield maximal independence between programs on the one hand and machine representation and organization of data on the other.

A further advantage of the relational view is that it forms a sound basis for treating derivability, redundancy, and consistency of relations—these are discussed in Section 2. The network model, on the other hand, has spawned a number of confusions, not the least of which is mistaking the derivation of connections for the derivation of relations (see remarks in Section 2 on the “connection trap”).

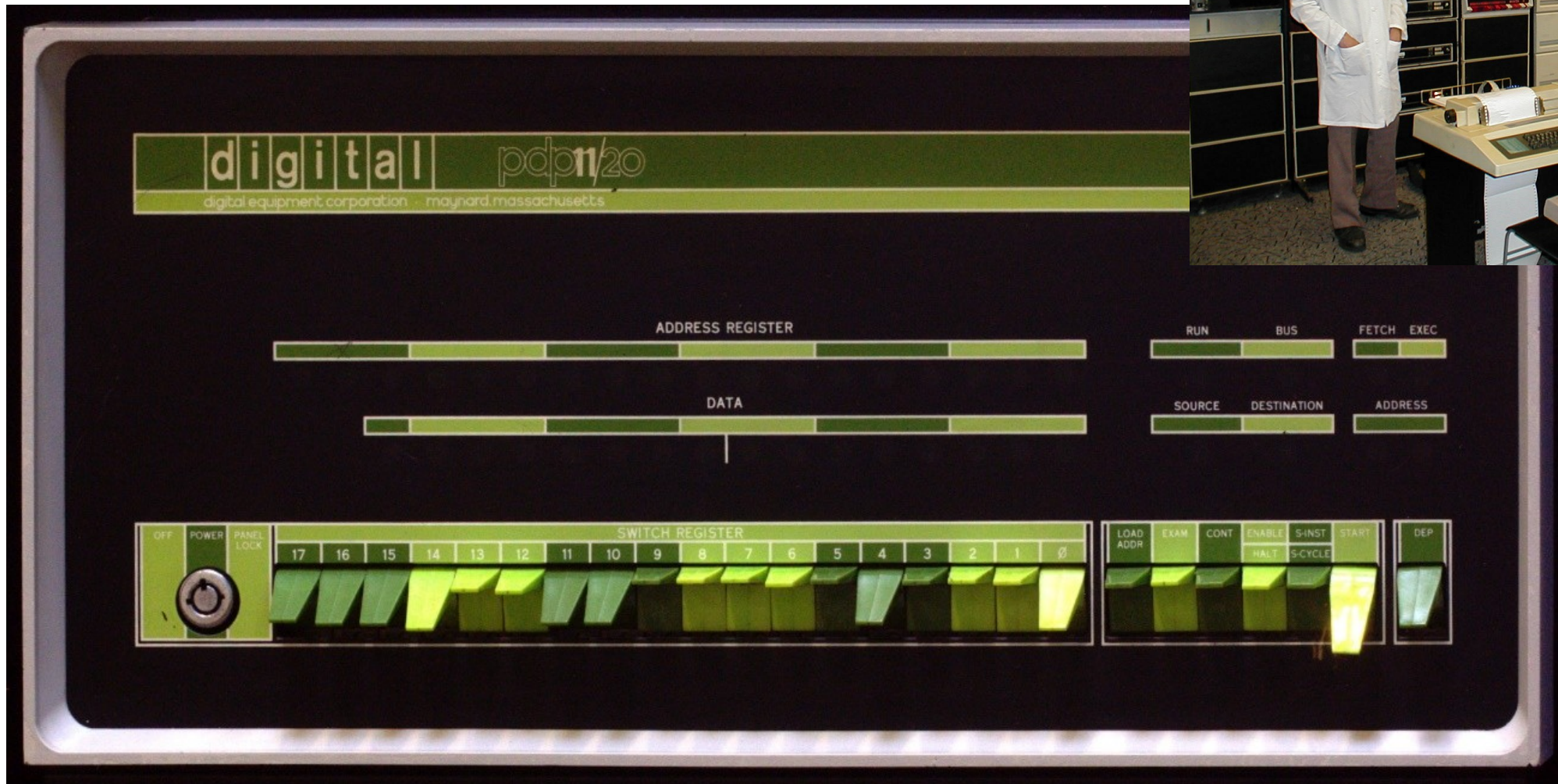
Finally, the relational view permits a clearer evaluation of the scope and logical limitations of present formatted data systems, and also the relative merits (from a logical standpoint) of competing representations of data within a single system. Examples of this clearer perspective are cited in various parts of this paper. Implementations of systems to support the relational model are not discussed.

1.2. DATA DEPENDENCIES IN PRESENT SYSTEMS

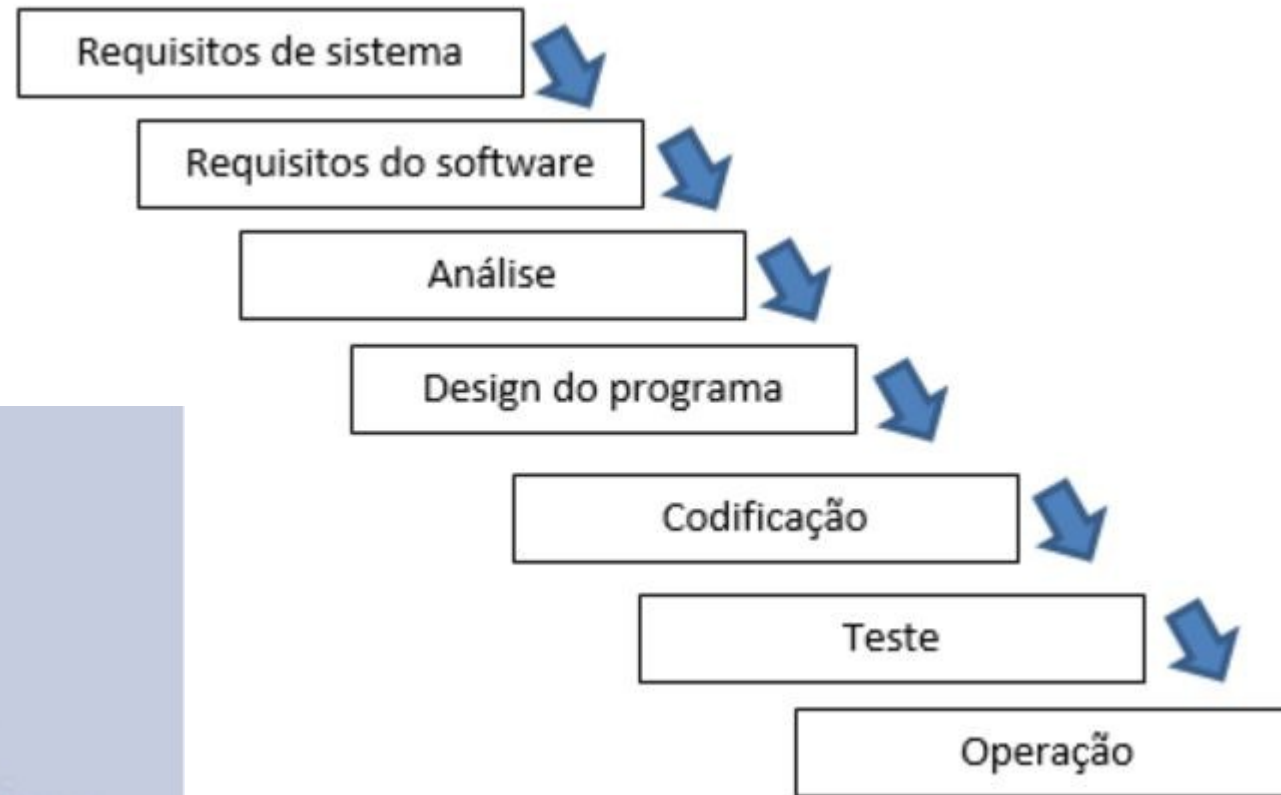
The provision of data description tables in recently developed information systems represents a major advance toward the goal of data independence [5, 6, 7]. Such tables facilitate changing certain characteristics of the data representation stored in a data bank. However, the variety of data representation characteristics which can be changed *without logically impairing some application programs* is still quite limited. Further, the model of data with which users interact is still cluttered with representational properties, particularly in regard to the representation of collections of data (as opposed to individual items). Three of the principal kinds of data dependencies which still need to be removed are: ordering dependence, indexing dependence, and access path dependence. In some systems these dependencies are not clearly separable from one another.

1.2.1. *Ordering Dependence.* Elements of data in a data bank may be stored in a variety of ways, some involving no concern for ordering, some permitting each element to participate in one ordering only, others permitting each element to participate in several orderings. Let us consider those existing systems which either require or permit data elements to be stored in at least one total ordering which is closely associated with the hardware-determined ordering of addresses. For example, the records of a file concerning parts might be stored in ascending order by part serial number. Such systems normally permit application programs to assume that the order of presentation of records from such a file is identical to (or is a subordering of) the

8.5 PDP-11 – 1970

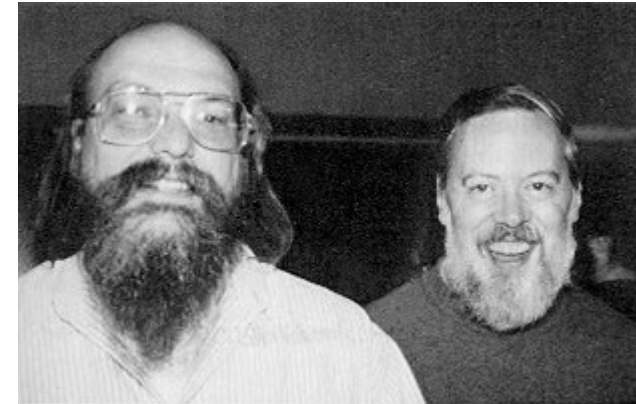


8.6 Modelo Cascata – 1970



Winston Royce (Estados Unidos, 1929-1995)

8.7 Unix – 1970

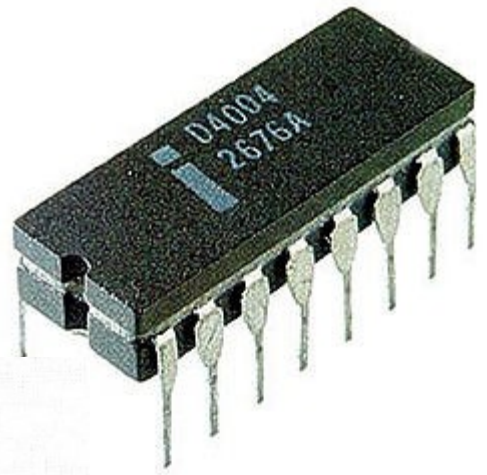


No beard, no belly, no guru

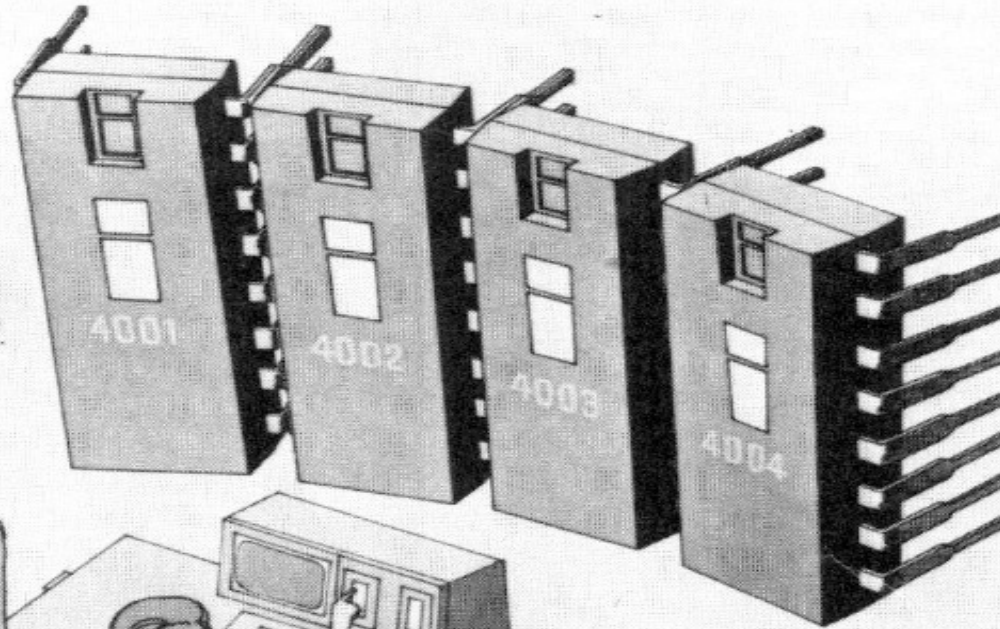
Kenneth Thompson (Estados Unidos, 1943).

Dennis Ritchie (Estados Unidos, 1941-2011)

8.8 INTEL 4004, o Primeiro Microprocessador – 1971



**Announcing
a new era
of integrated
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A micro- programmable computer on a chip!

Intel introduces an integrated CPU complete with a 4-bit parallel adder, sixteen 4-bit registers, an accumulator and a push-down stack on one chip. It's one of a family of four new ICs which comprise the MCS-4 micro computer system—the first system to bring you the power and flexibility of a dedicated general-purpose computer at low cost in as few as two dual in-line packages.

MCS-4 systems provide complete computing and control functions for test systems, data terminals, billing machines, measuring systems, numeric control systems and process control systems.

The heart of any MCS-4 system is a Type 4004 CPU, which includes a powerful set of 45 instructions. Adding one or more Type 4001 ROMs for program storage and data tables gives you a fully functioning micro-programmed computer. To this you may add Type 4002 RAMs for read-write memory and Type 4003 registers to expand the output ports.

Using no circuitry other than ICs from this family of four, you can create a system with 4096 8-bit bytes of ROM storage and 5120 bits of RAM storage. When you require rapid turn-around or need only a few systems, Intel's erasable and re-programmable ROM, Type 1701, may be substituted for the Type 4001 mask-programmed ROM.

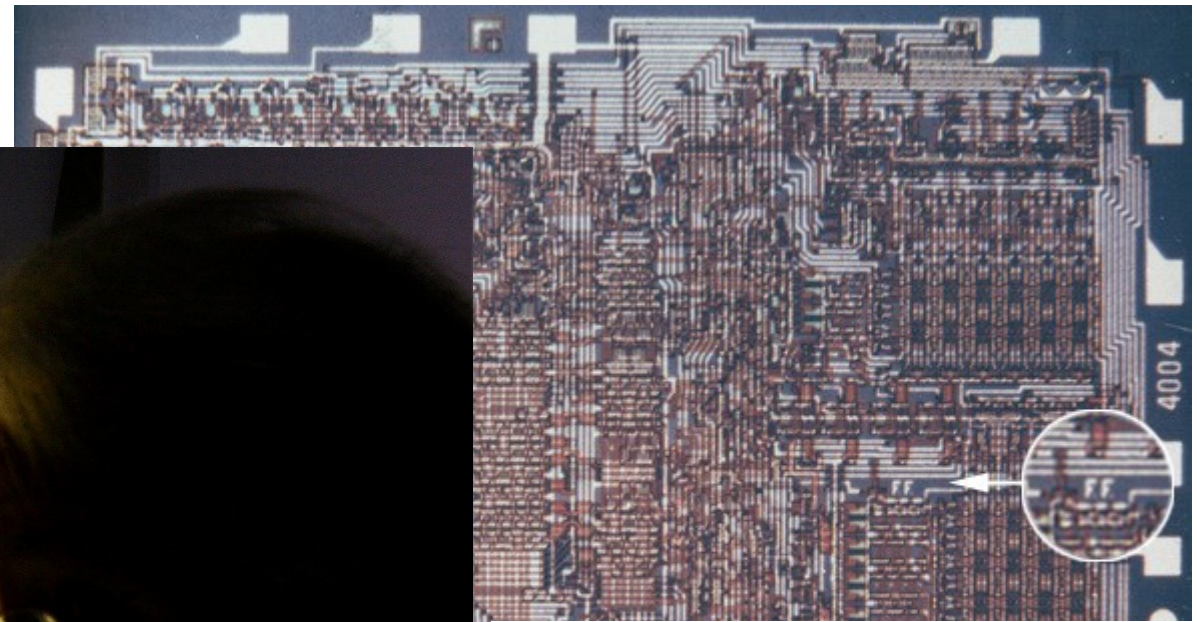
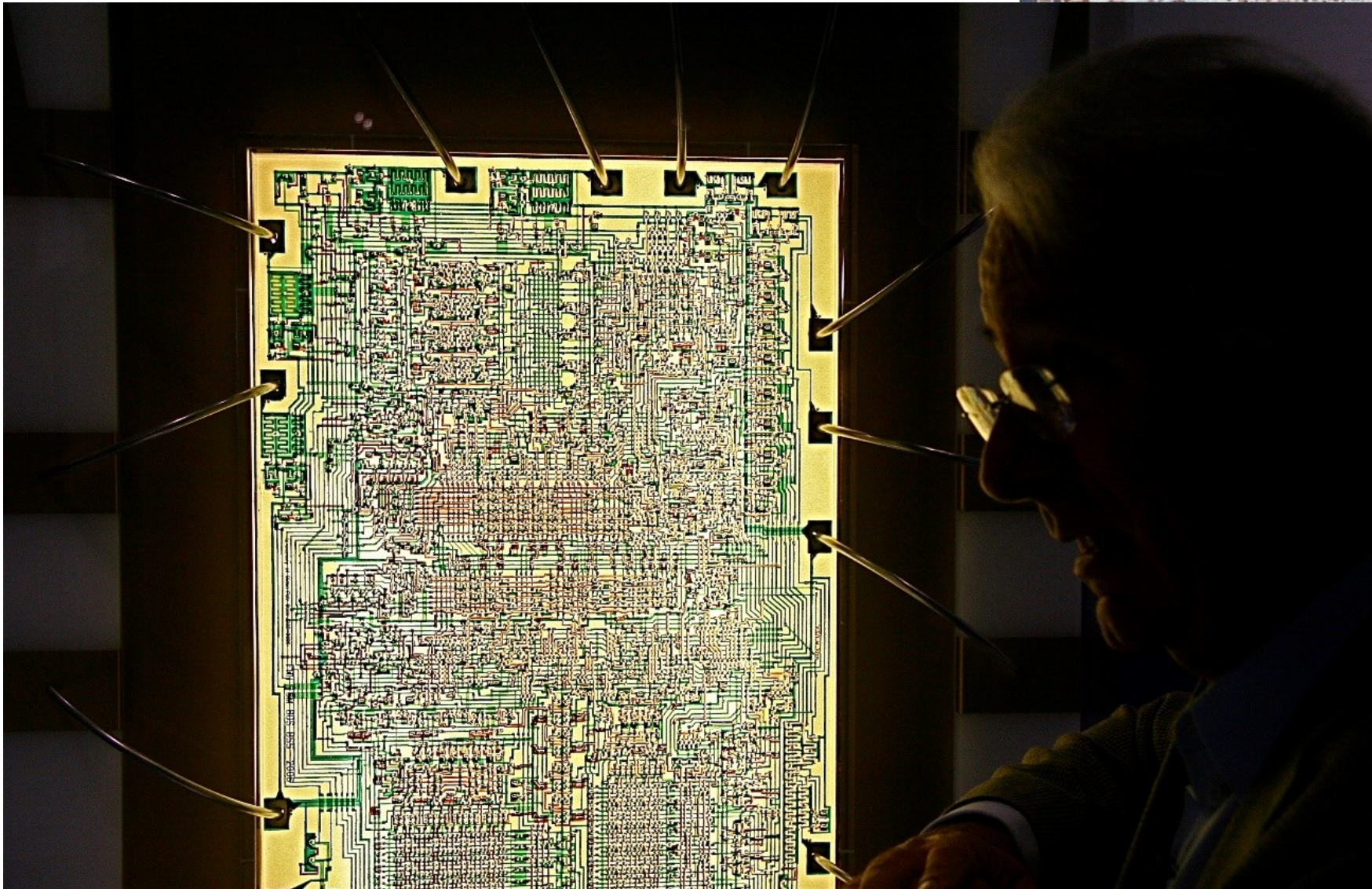
MCS-4 systems interface easily with switches, key-boards, displays, teletypewriters, printers, readers, A-D converters and other popular peripherals.

The MCS-4 family is now in stock at Intel's Santa Clara headquarters and at our marketing headquarters in Europe and Japan. In the U.S., contact your local Intel representative for technical information and literature. In Europe, contact Intel at Avenue Louise 218, B-1050 Bruxelles, Belgium. Phone 492063. In Japan, contact Intel Japan, Inc., Parkside Flat Bldg. No. 9-2-2, Sendagaya, Shibuya-Ku, Tokyo 151. Phone 03-403-4747.

Intel Corporation now produces micro computers, memory devices and memory systems at 3065 Bowers Avenue, Santa Clara, Calif. 95051. Phone (408) 246-7501.

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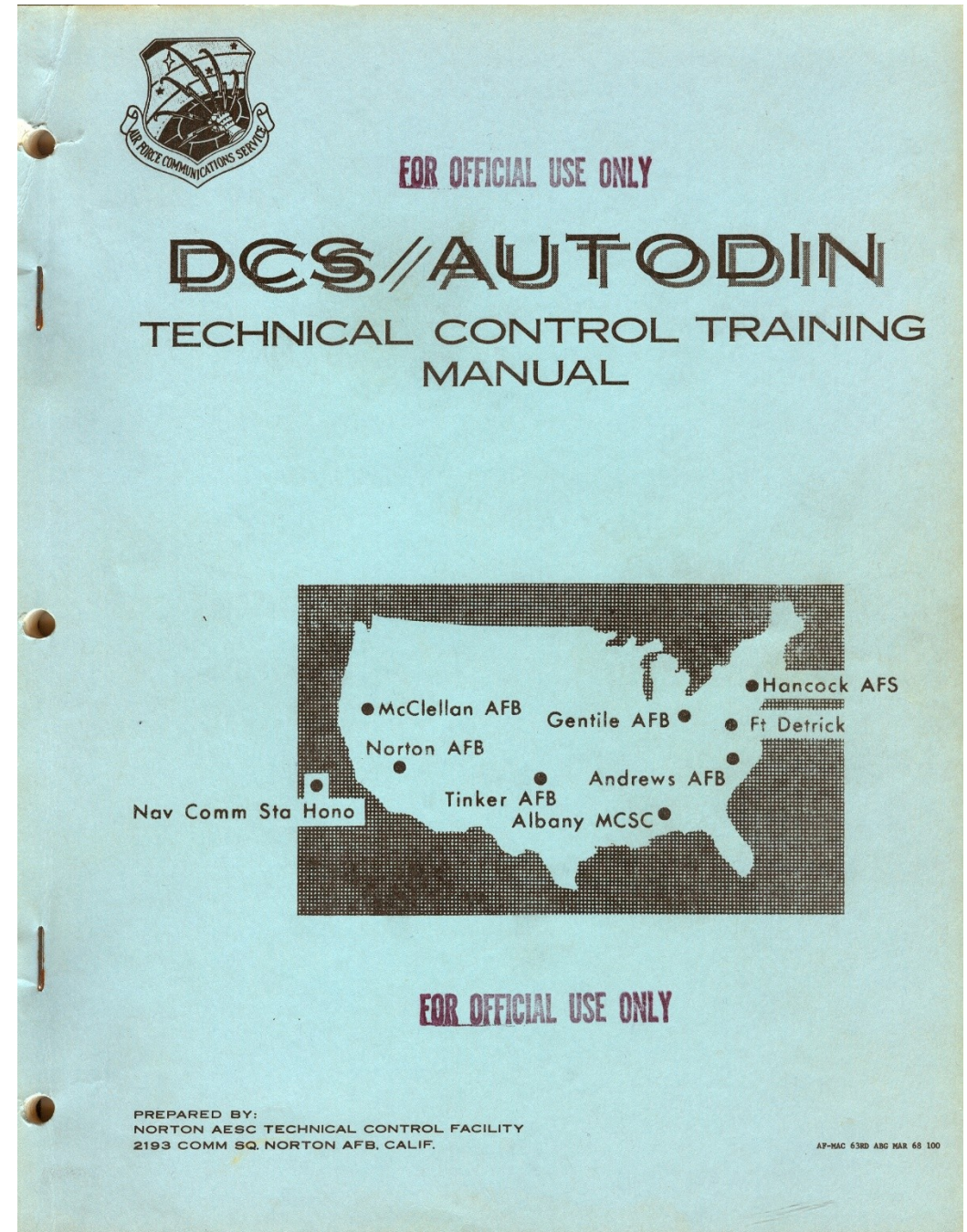


Federico Faggin (Itália, 1941),

8.9 E-Mail – 1971



Ray Tomlinson (Estados Unidos, 1941)



8.10 Projeto Gutenberg, a Primeira Biblioteca Digital – 1971



Michael S. Hart (Estados Unidos, 1947-2011)

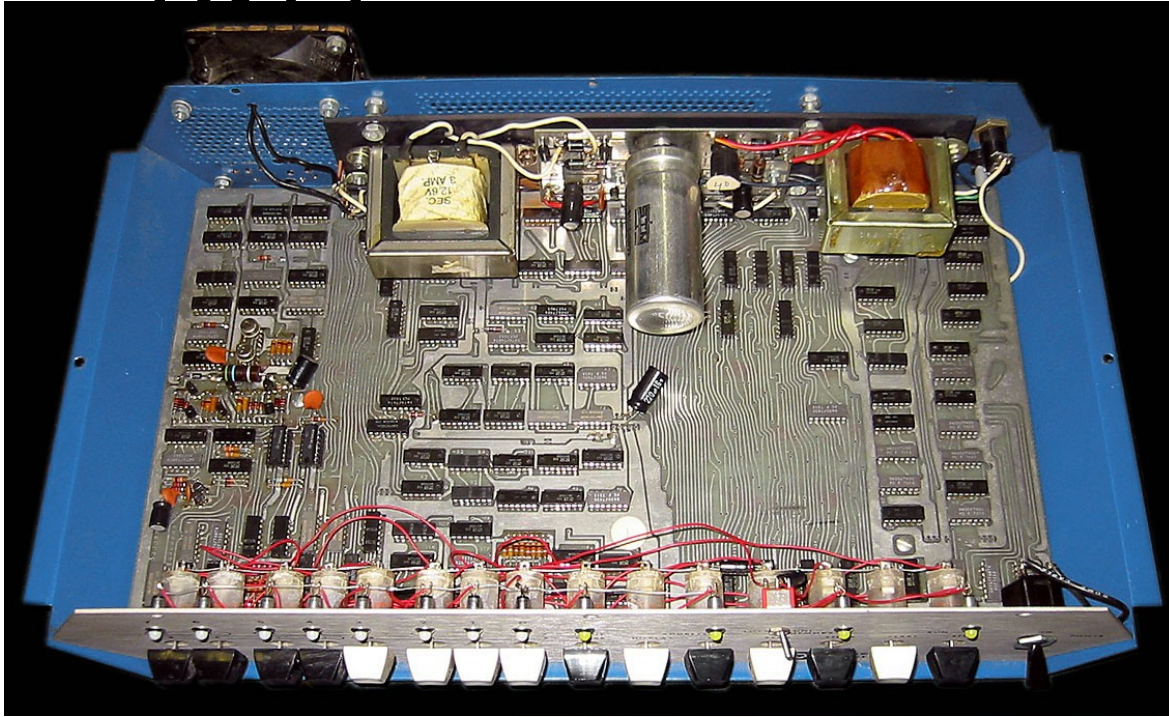


8.11 The Creeper, o Primeiro Vírus – 1971



IN THE CREEPER. CATCH ME IF YOU CAN!

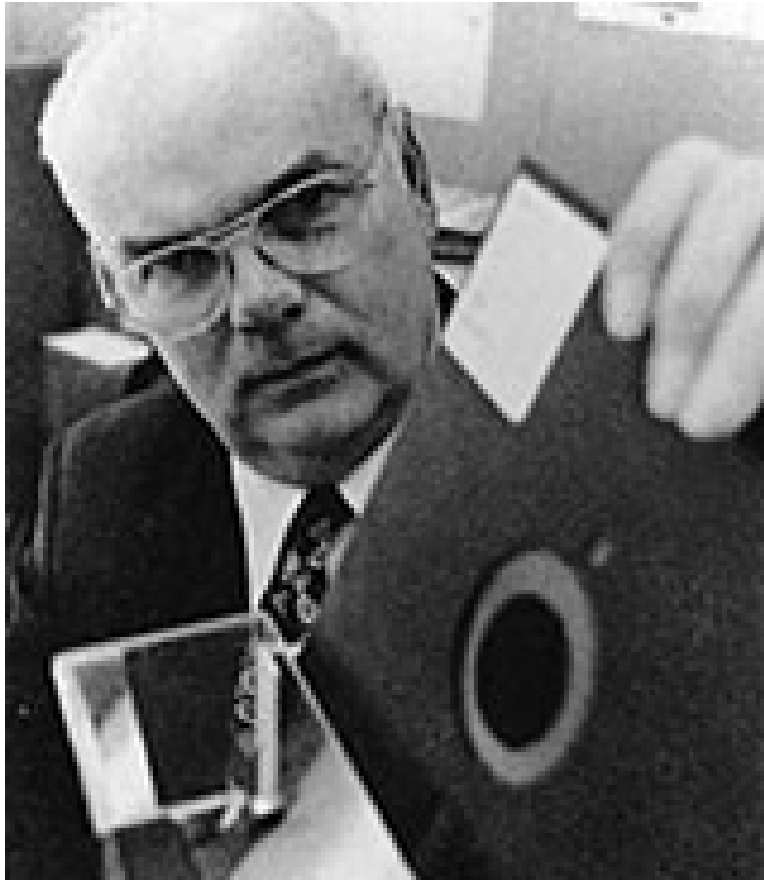
8.12 Computador Pessoal Kenbak-1 – 1971



John Blankenbaker



8.13 Disco Flexível – 1977



David L. Noble

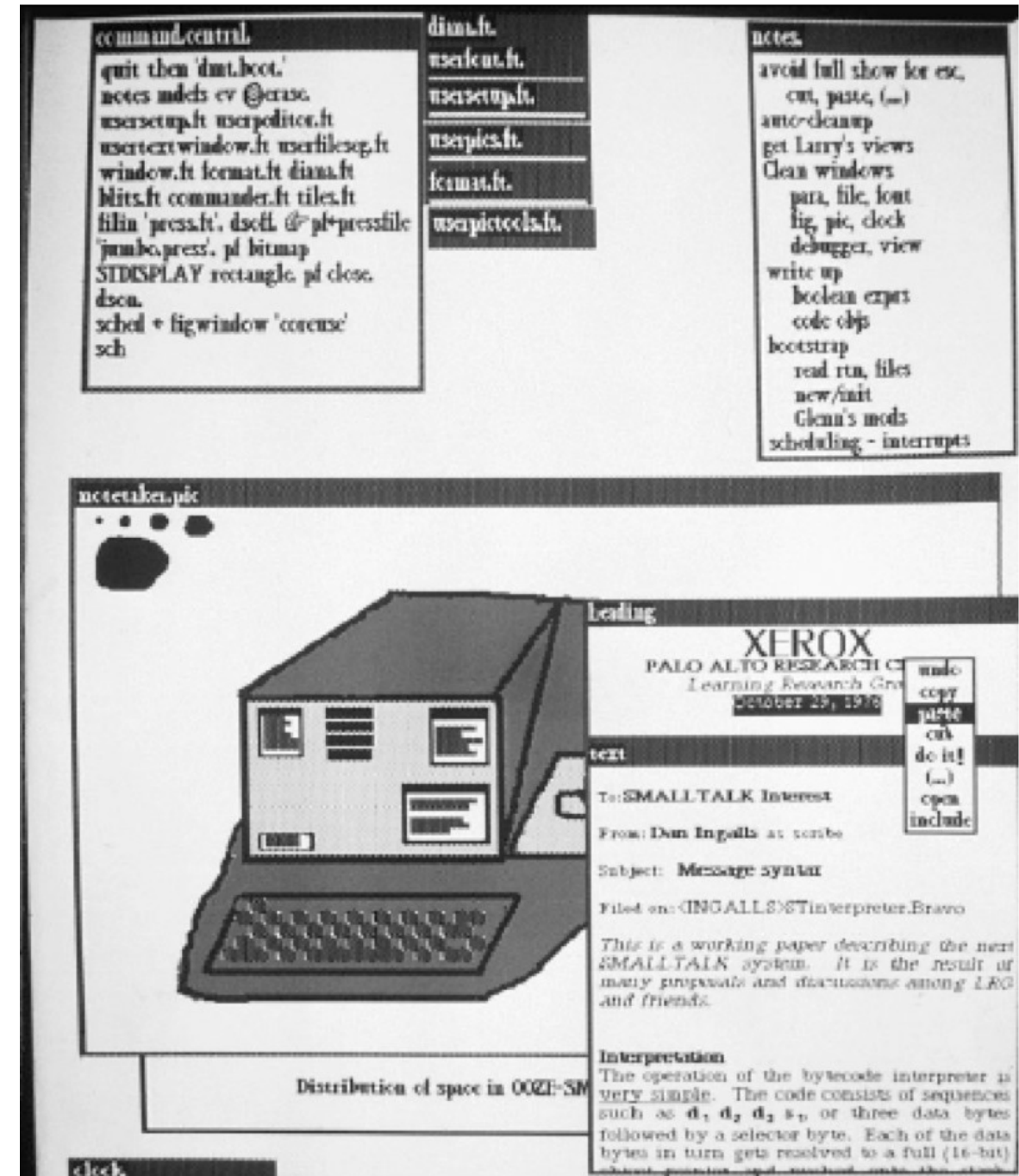


8.14 Smalltalk – 1971

```
4 timesRepeat: [turtle forward: 10; turnRight: 90]
```



Adele Goldberg (Estados Unidos, 1945)



8.15 C – 1972



Ken Thompson (Estados Unidos, 1943)

Dennis Ritchie (Estados Unidos, 1941-2011)

```
#include <math.h>
#include <sys/time.h>
#include <X11/Xlib.h>
#include <X11/keysym.h>

double L, o, P,
    _dt, T, Z, D=1, d,
    s[999], E, h= 8, I,
    J, K, w[999], M, m, O,
    n[999], j=33e-3, i=
    1E3, r, t, u, v, W, S=
    74.5, l=221, X=7.26,
    a, B, A=32.2, c, F, H;
int N, q, C, y, p, U;
Window z; char f[52];
; GC k; main(){ Display *e=
XOpenDisplay( 0); z=RootWindow(e,0); for (XSetForeground(e,k=XCreateGC(e,z,0,0),BlackPixel(e,0))
; scanf("%lf%lf%lf",y+n,w+y,y+s)+1; y++; XSelectInput(e,z= XCreateSimpleWindow(e,z,0,0,400,400,
0,0,WhitePixel(e,0)),KeyPressMask); for(XMapWindow(e,z); ; T=sin(O)){ struct timeval G={ 0,dt*1e6}
; K= cos(j); N=1e4; M+= H*_; Z=D*K; F+= *_P; r=E*K; W=cos( O); m=K*W; H=K*T; O+=D*_F/ K+d/K*_E; B=
sin(j); a=B*T*D-E*W; XClearWindow(e,z); t=T*_E+ D*B*_W; j+=d*_D*_F*_E; P=W*_E*B-T*D; for (o+=(I-D*W*_E
*T*B,E*d/K *_B+v+B/K*_F*D)*_; p<y; ){ T=p[s]+i; E=c-p[w]; D=n[p]-L; K=D*m-B*T-H*_E; if(p [n][w][ p]+p[s
]-- 0|K <fabs(W-T*r-I*_E +D*P) |fabs(D-t *_D+Z *_T-a *_E)> K)N=1e4; else{ q=W/K *_4E2+2e2; C= 2E2+4e2/ K
*_D; N=1E4&& XDrawLine(e ,z,k,N ,U,q,C); N=q; U=C; } ++p; } L+=_ (X*t +P*M+m*1); T=X*X+ 1*1+M *_M;
XDrawString(e,z,k ,20,380,f,17); D=v/1*15; i+=(B *_1-M*r -X*Z)*_; for(; XPending(e); u *_CS!-N){
XEvent z; XNextEvent(e ,&z);
++*((N=XLookupKeysym
(&z.xkey,0))-IT?
N-LT? UP-N?& E:&
J:& u: &h); --*(
DN -N? N-DT ?N--
RT?&u: & W:&h:&J
); } m=15*_F/1;
c+=(I=M/ 1,1*_H
+I*M+a*X)*_; H
=A*r+v*X-F*1+(
E-.1+X*4.9/1,t
=T*m/32-I*T/24
)/S; K=F*M+(
h*_1e4/1-(T+
E*5*T*_E)/3e2
)/S-X*d-B*A;
a=2.63 /1*d;
X+=( d*_1-T/S
*(.19*_E +a
*.64+J/1e3
)-M*_v +A*_
Z)*_; l +=
K *_; W=d;
sprintf(f,
"%5d %3d"
"%7d",p,-l
/1.7,(C-9E3+
O*57.3)%0550,(int)i); d+=T*(.45-14/1*
X-a*130-J*_14)*_/_125e2+F*_*_v; P=(T*(47
*I-m*_52+E*94 *_D-t*.38+u*.21*_E) /1e2+W*
179*v)/2312; select(p=0,0,0,0,&G); v-=
W*_F-T*(.63*m-I*.086+m*_E*19-D*25-.11*u
)/107e2)*_; D=cos(o); E=sin(o); } }
```

8.16 PROLOG – 1972

Uma expressão PROLOG, que define que gatos comem ratos poderia então ser descrita assim:

```
come(x,y) :- gato(x), rato(y).
```

Dessa forma, se afirmarmos:

```
gato(tom).  
rato(jerry).  
come(tom,?).
```

A resposta será:

```
jerry
```

Por outro lado, se trocarmos a terceira expressão por:

```
come(?,jerry)
```

A resposta será:

```
tom
```



Alain Colmerauer (França, 1941)

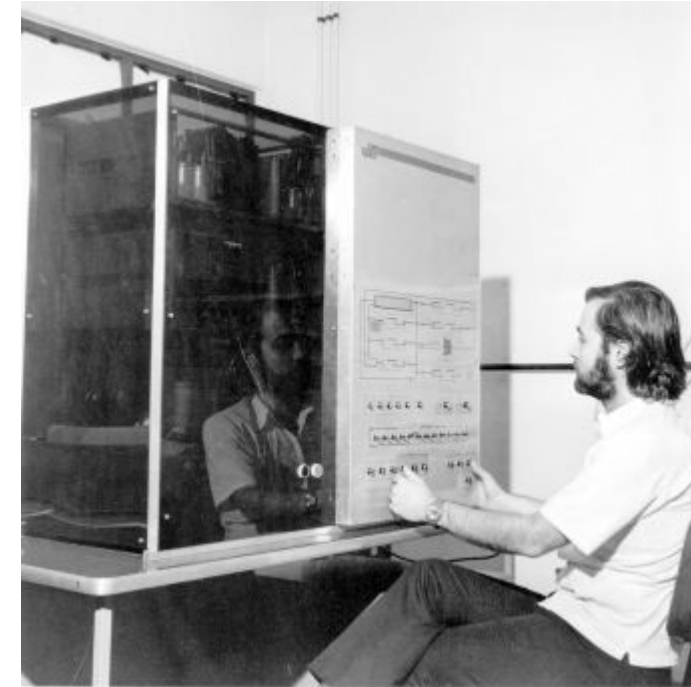
Usando essa linguagem, Colmerauer e Roussel implementaram um sistema de comunicação homem-máquina em 1972, com 610 cláusulas (definições). Segue um exemplo (traduzido) de diálogo entre homem e máquina obtido por eles com este sistema. Inicialmente eles apresentam as regras, fatos e questões:

```
Todo psiquiatra é uma pessoa.  
Toda pessoa que ele analisa está doente.  
Jacques é um psiquiatra em Marseille.  
Jacques é uma pessoa?  
Onde está Jacques?  
Jacques está doente?
```

As respostas obtidas do computador foram:

```
Sim.  
Em Marseille.  
Eu não sei.
```


8.17 Pato Feio – 1972

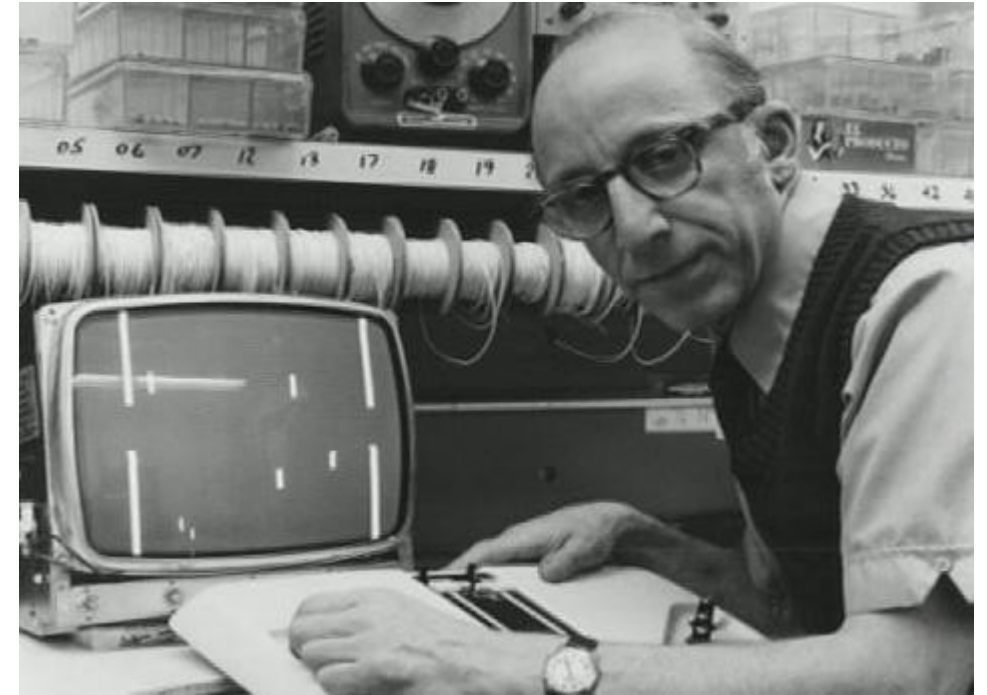


Antonio Hélio Guerra Vieira

8.18 Magnavox Odyssey – 1972



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Ralph H. Baer (Alemanha, 1922-2014)

8.19 Calculadora HP 9830A – 1972

HP 9830A/B* USERS.

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Where more memory is important, the user area may be increased to 32K bytes with our EM-30...the first and only 16,096-word memory for the 9830A/B! Compared to newly introduced models equipped with commonly used ROM options, the 9830A/B with the EM-30 provides 8,192 bytes more user memory. A truly significant increase in capacity!

With the 9830A/B, the EM-30 provides 2,048 more bytes of memory, is priced right, and is fully warranted for a year.

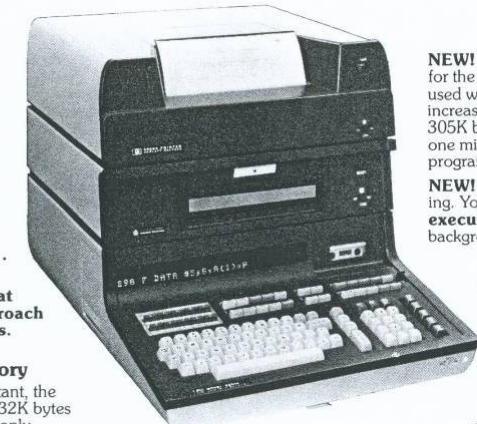
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Infotek's FAST BASIC ROMS add to the machine instruction set where HP left off. These ROMS provide spectacular increases in the work throughput of your 9830A/B. For example, you can process arrays at speeds of 40,000 words per second, attain an I/O capability of 10,000 bytes per second, greatly increase the power of a 9880A/B disk system, and print from a buffer while computing.

By providing 46 additional functions, statements, and commands, the FAST BASIC ROMS give your 9830A/B the most powerful BASIC language repertoire short of \$200,000 systems.

If you need FAST BASIC but don't have enough unused ROM slots, don't give up! You can install our FAST BASIC ROMS without losing any present capability. **Contact us, we'll show you!**

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HP 9830A/B with the Infotek FD-30 Mass Memory

Floppy Disk Memory System

The FD-30A provides 305K bytes of on-line data that can be searched 50 times faster than your present cassette system. Data throughput is actually four times faster than the 9880A/B Mass Memory. Best of all, no software modifications are required since the FD-30A obeys all cassette syntax.

The FD-30A is the optimum magnetic memory for the 9830A/B. It has the right capacity, speed, and price.

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Less trade-in of HP 8K Words	1,100	
FAST BASIC I ROM	675	
FAST BASIC II ROM	525	
FAST BASIC I & II as a set	1,050	
FAST BASIC III ROM	525	
FD-30A Mass Memory	3,895	
FD-30S Slave Drive	1,780	
FD-30D Dual Drive System	4,900	
RS-30 High Speed RS 232C Interface	750	
TC-30 Real Time Clock	825	
LP-30 Heavy Duty Line Printer	6,850	
CP-30 Character Printer	3,400	
PS-30 Paper Tape Punch/Reader	3,850	
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NEW! The FD-30S low cost Slave Drive for the FD-30A. Multiple slaves can be used with a single FD-30A so you can increase on-line capacity in blocks of 305K bytes. Also, the FD-30S provides one minute disk duplication under program control or stand-alone.

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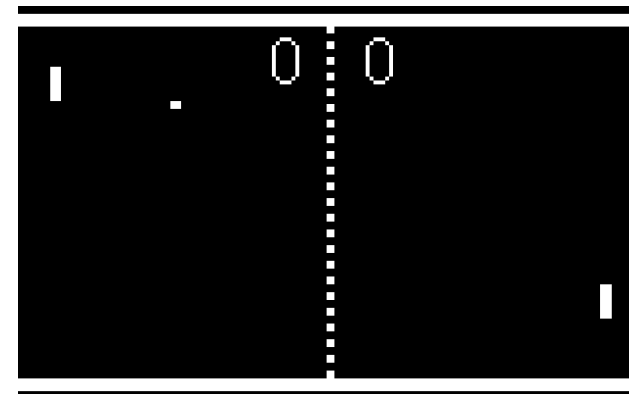
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8.20 Pong – 1972



Nolan Bushnell (Estados Unidos, 1943) e Ted Dabney

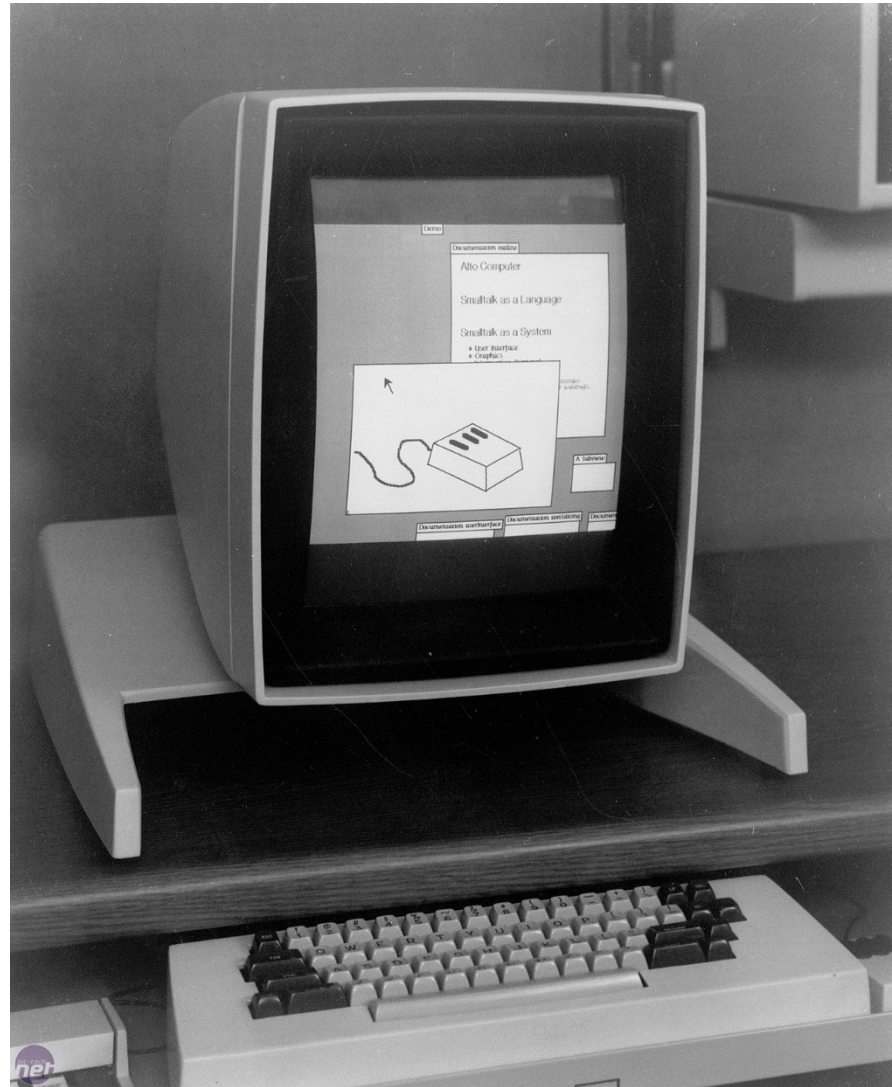


8.21 Ethernet – 1973



Robert Metcalfe (Estados Unidos, 1946)

8.22 Xerox Alto – 1973



8.23 Micral – 1973



8.24 Salas de Chat Talkomatic – 1973

```
TALK (Channel 1)

$$$ Dr. Woolley / cerl 1-20
    Okay, that's better.
    I was hoping Papa Del's is still around - it was the best

$$$ brian / uofdel 1-25
    is garcia's still in business?

$$$ Δ / s 5-16
    I haven't eaten at delz for a while
    it's better than garcias tho.

$$$ Loren / unl 1-18
    *grin* and *sigh*

$$$ Doug / nginear 1-0
    hmmm
```



David R. Woolley

8.25 Telefone Celular Pessoal – 1973



Martin Cooper (Estados Unidos, 1928)

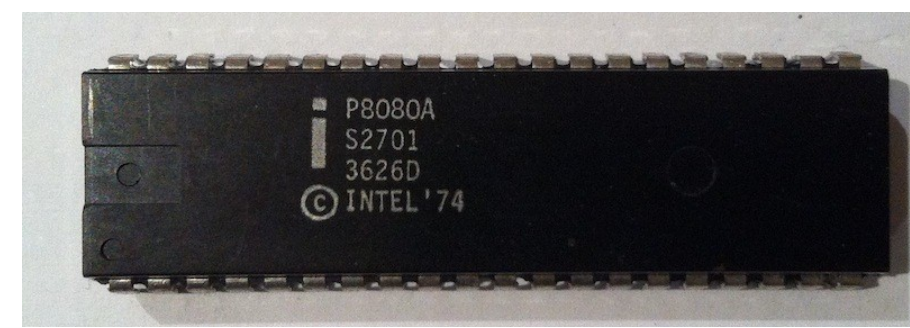


Martin Cooper (On Right) Making The World's First Cellular Phone Call To Joel S. Engel (On Left)

8.26 Community Memory, o Primeiro BBS – 1973



8.27 Intel 8080 – 1974



Federico Faggin (1972)

From CPU 8080 Microcom

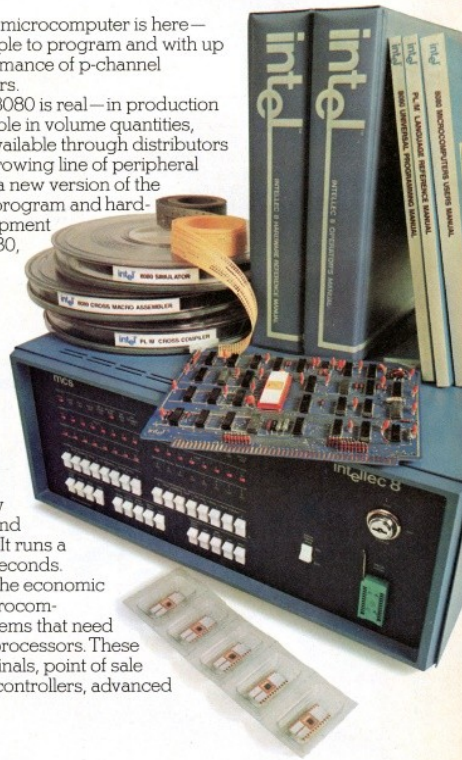
Intel's new 8080 n-channel microcomputer is here—
incredibly easy to interface, simple to program and with up
to 100 times the performance of p-channel
MOS microcomputers.

Best of all, the 8080 is real—in production
at Intel and available in volume quantities,
today. It's also available through distributors
along with a growing line of peripheral
circuits and a new version of the
Intellec 8, a program and hard-
ware development
system for the 8080,

all supported with software
packages, design documenta-
tion and manuals, and backed
by more than 100 man years
of microcomputer expertise.

The 8080 is the inevitable
successor to complex custom
MOS and many large discrete
logic subsystems. It is the
industry's first general purpose
n-channel microcomputer and
the first high performance
single-chip CPU, with extremely
simple interface requirements and
straightforward programming. It runs a
full instruction cycle in 2 microseconds.

As such, the 8080 extends the economic
benefits of Intel's p-channel microcom-
puters to a new universe of systems that need
fast, multi-port controllers and processors. These
systems include intelligent terminals, point of sale
systems, process and numeric controllers, advanced



to software, the puter is here.

calculators, word processors, self-calibrating instruments, data loggers, communica-
tions controllers, and many more.

You can use 256 input and 256
output channels, handle almost
unlimited interrupt levels, directly
access 64 kilobytes of memory,
and put many satellite 8080 proc-
essors around a single memory.

Interfacing is minimal and
design is easy with the 8080
because all controls are fully
decoded on the CPU chip
itself and inputs and outputs are TTL
compatible. There are separate data,
address and control buses.

The 8080 microcomputer has
78 basic instructions, including the
8008 set plus new ones that make possible such features as vectored multi-level
interrupt, unlimited subroutine nesting and very fast decimal and binary
arithmetic.

Program development for the 8080 can be done either on a large
computer using the Intel software cross products (PL/M systems language
compiler, macro-assembler and simulator), or on an Intellec 8 development
system with a resident monitor, text editor and macro-assembler.

The new 8080 product family includes performance matched peripheral and
memory circuits configured to minimize design effort and maximize system
performance. Large, low cost RAMs, ROMs, PROMs and I/O devices are available
now and we will soon announce other 8080 LSI support circuits.

The 8080 is easier to use and more economical than any high performance
microcomputer in sight. It's here now, in volume, from the inventors of the micro-
computer and the people who lead the industry in production and design support.
Intel Corporation, 3065 Bowers Avenue, Santa Clara, California 95051.
(408) 246-7501.

intel Microcomputers. First from the beginning.

RAM MEMORIES			
8101	256 x 4 STATIC	AVL:	3RD Q
8111	256 x 4 STATIC WITH COMMON I/O	AVL:	3RD Q
8102	1024 x 1 STATIC	AVL:	NOW
8107	4096 x 1 DYNAMIC	AVL:	NOW
ROM MEMORIES			
8308	1024 x 8	AVL:	3RD Q
8316	2048 x 8	AVL:	NOW
PROM MEMORIES			
8702	256 x 8	AVL:	NOW
8704	512 x 8	AVL:	4TH Q
8604	512 x 8	AVL:	4TH Q
PERIPHERAL CIRCUITS			
8205	1 of 8 DECODER	AVL:	NOW
8210	4K RAM CLOCK DRIVER	AVL:	NOW
8212	8-BIT I/O LATCH	AVL:	NOW
8216	8I-DIRECTIONAL BUS DRIVER	AVL:	3RD Q
8201	COMMUNICATIONS INTERFACE	AVL:	4TH Q

INTEL 8080 PRODUCT FAMILY

8.28 Altair 8800 – 1974



Henry Edwards Roberts (Estados Unidos, 1941-2010)

HOW TO "READ" FM TUNER SPECIFICATIONS

Popular Electronics

WORLD'S LARGEST-SELLING ELECTRONICS MAGAZINE JANUARY 1975/754

PROJECT BREAKTHROUGH!

World's First Minicomputer Kit to Rival Commercial Models...

"ALTAIR 8800" SAVE OVER \$1000



ALSO IN THIS ISSUE:

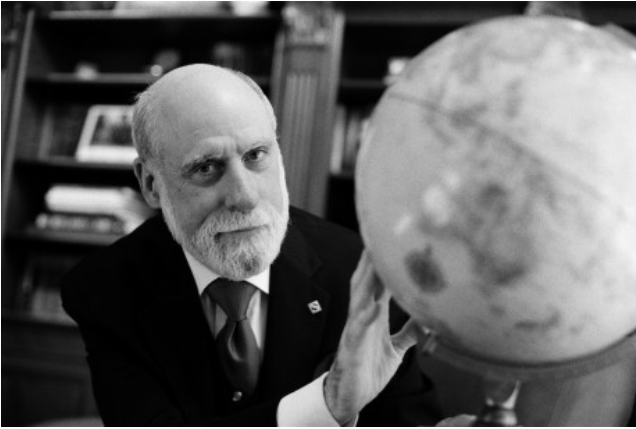
- An Under-\$90 Scientific Calculator Project
- CCD's—TV Camera Tube Successor?
- Thyristor-Controlled Photoflashers

TEST REPORTS:

- Technics 200 Speaker System
- Pioneer RT-1011 Open-Reel Recorder
- Tram Diamond-40 CB AM Transceiver
- Edmund Scientific "Kirlian" Photo Kit
- Hewlett-Packard 5381 Frequency Counter



8.29 TCP-IP e o Surgimento da Internet – 1974

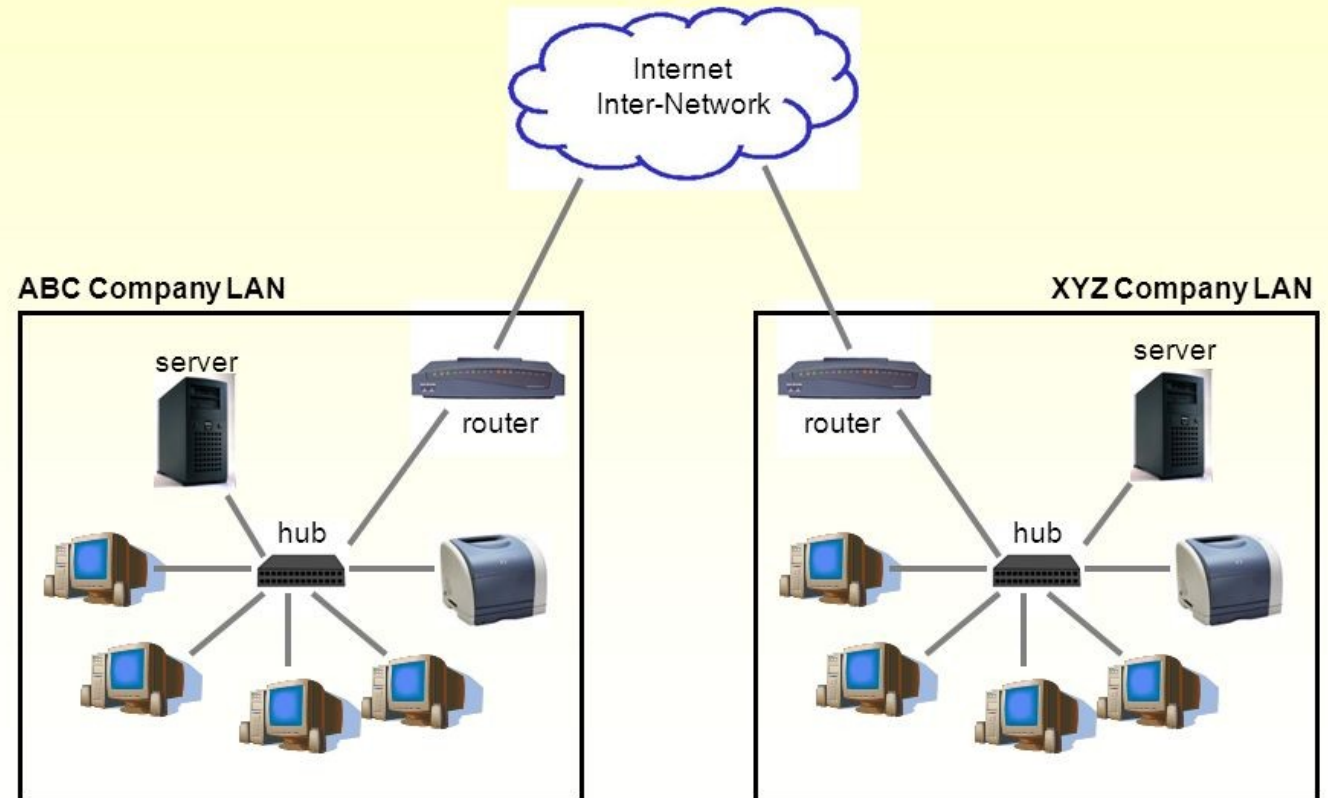


Vinton Gray Cerf (Estados Unidos, 1943)



Robert Elliot Kahn (Estados Unidos, 1938)

The Internet – “Inter-Network”
TCP/IP Protocol allows LANs to be Interconnected



8.30 COBRA, Computadores Brasileiros



UFRPE – Universidade Federal Rural de Pernambuco



O primeiro computador Cobra no Brasil



Cobra 700, primeiro computador lançado pela Cobra.

A Cobra ainda em meados de 1976, que na mesma época que absorvia todos os técnicos do Laboratório de Projetos em Computação da PUC-Rio, que a universidade acabava de extinguir.

Os bancos haviam sido "empurrados" para a associação com a Cobra pelo BNDE e pela Capre, que desestimularam a formação de joint-ventures entre bancos como o Bradesco e multinacionais.

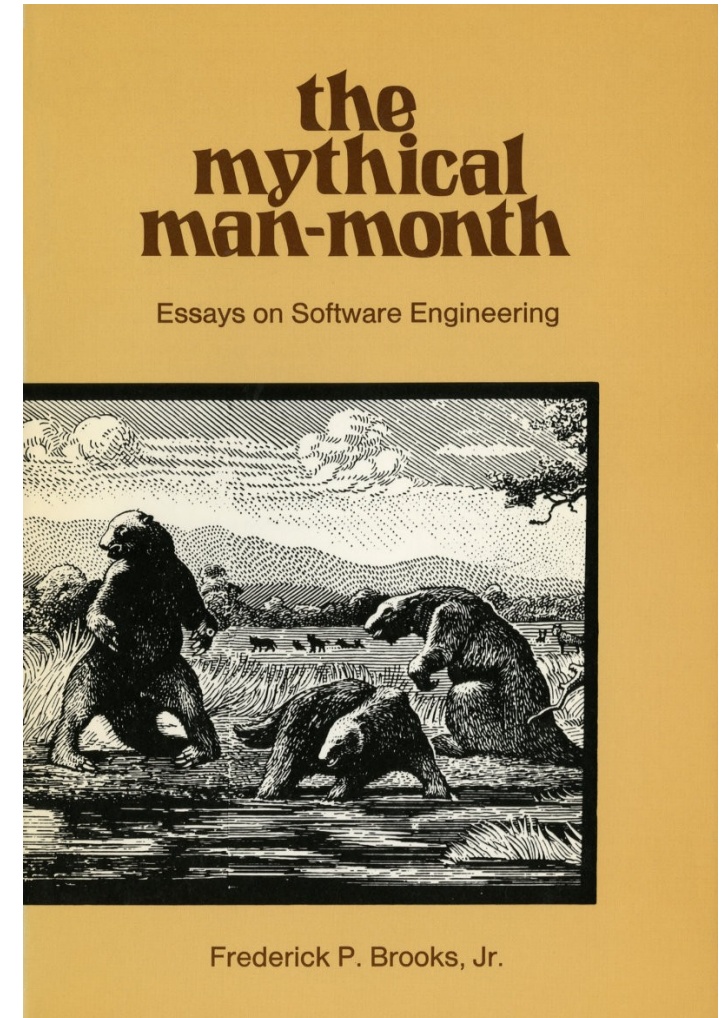
8.31 The Mythical Man-Month – 1975



You can learn more from failure than success. In failure you're forced to find out what part did not work. But in success you can believe everything you did was great, when in fact some parts may not have worked at all. Failure forces you to face reality.

— Fred Brooks —

AZ QUOTES



8.32
1975

Computador Portátil IBM 5100 –



IBM announces the new 5100 Portable Computer

A compact problem-solving aid for engineers,
statisticians, scientists and financial and business analysts.

Now you can have a computer right on your desk. Exactly where you need it. When you need it.

The new IBM 5100 Portable Computer incorporates the latest in semi-conductor technology. It features a typewriter-like keyboard and numeric key-pad for simplified data entry, a 1024 character display screen, an integrated magnetic tape drive, and 16K characters of memory.

Options available with the 5100 include a bi-directional 80-characters per second printer, a second magnetic tape drive, and additional memory up to a maximum of 64K characters. Also available is a communications feature which allows the 5100 to be used as a terminal.

The IBM 5100 comes with either APL or BASIC language or both.

Over 100 often-used analytical routines in mathematical, statistical and financial calculations are available for such functions as forecasting, modeling, matrix arithmetic, engineering and design calculations, regression and correlation analysis, return on investment and cash flow analysis.

In addition, the 5100 features a self-study training package that makes it easy to learn and easy to use without taking any classes or relying on specially trained experts.

If you'd like to find out more about IBM's new 5100 Portable Computer and arrange for a demonstration right at your desk, call your IBM General Systems Division office or fill out this coupon.

IBM	IBM General Systems Division P.O. Box 2068, Atlanta, Georgia 30301	SA
☐ I would like more information about IBM's new 5100.		
☐ I would like a demonstration of IBM's new 5100.		
My major area of interest is:		
☐ Engineering/Scientific ☐ Statistical Analysis		
☐ Business/Financial Analysis		
Name _____		
Title _____		
Company _____		
Address _____		
City _____ State _____ Zip _____		
Phone _____		

8.33 Sphere – 1975

Go Computer Now!
Why not?

FROM \$860 TO \$11,300 SPHERE CAN'T BE BEAT!

SPHERE starts with a CPU using a Motorola 6800 microprocessor, a Real-Time Clock, 4K of dynamic memory, 1K of PROM software. The CRT Board generates 16 lines by 32 characters of ASCII on a television or video monitor. Keyboard is complete with numeric and cursor editing keypads. From here, hardware can be expanded to your hearts desire with extra memory boards (up to 64K), serial communications interface, cassette interface, Modem, digital I/O (as many as you need). Floppy Disk memory (up to 4 disks), 8 computer terminals, line printer etc...all from one M6800 chip.

With a SPHERE Computer, stand-alone development is just the beginning, you can configure your system to handle your problem solving/record keeping needs. All SPHERE Computer Systems come complete with useable software languages. Available are "PDS" 1K Basic, or extended Basic Compiler. When computer is turned on, it immediately goes into a comm- and mode, so that you can instantly start programming. "PDS" contains a mini-assembler, editor, debugger, and utility comm- and set in 1K of PROM. Also available is a 1K subset of Basic. Our extended Basic compiler is complete with string, matrix, and file functions, and requires 12K of memory. With this software you can perform your applications whether it be accounting, home management, education, security monitoring, research, business, etc. Why not invent your own application? For play or for work your biggest problem is no problem at all. **\$860 is the start for an operating Computer System Kit.** Your computer is ready and comes complete with operator manuals sufficient for first-time computer users. Contact us today for more information.

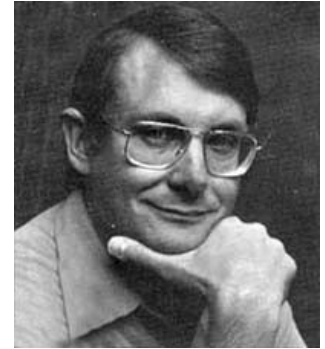
KIT	ASM		KIT	ASM	
\$350	\$520	ONE-CARD COMPUTER: Motorola 6800 microprocessor, 4K RAM, 512 bytes EPROM (containing a Program Development System), a REAL-TIME CLOCK, 16 LINES OF DIGITAL I/O, hard wired ROM Monitor, and a serial type interface. This is the 100-quantity price, extended to the hobby user for a limited time on a single unit.	\$999	\$1499	SPHERE 2: Includes all features of SPHERE 1, plus serial communications and audio cassette or MODEM interface.
522	622	CPU BOARD: Motorola 6800 microprocessor, 4K RAM, 1K EPROM (containing an EDITOR, ASSEMBLER, DEBUGGER, COMMAND LANGUAGE, CASSETTE LOADER, DUMPER, UTILITIES), and a REAL-TIME CLOCK.	1765	2250	SPHERE 3: Includes all the features of SPHERE 2, plus memory totaling 20K which is sufficient to run full extended BASIC Language.
860	1400*	SPHERE 1: Includes the CPU BOARD described above, plus 512 character video with full ASCII keyboard and numeric/cursor keypad, power supply, chassis, manuals and associated parts.	6100	7995	SPHERE 4: Includes all of the features of SPHERE 3, except the cassette has been replaced by an IBM-compatible Dual Floppy Disk System. This system includes a Disk-operating System and BASIC Language and a 65 LPM line printer.
		(various)			OTHER SPHERE PRODUCTS: Light pen option; full color and B/W video graphics system; low cost Dual Floppy Disk System; and full line of low cost peripherals.

*This ASSEMBLED SPHERE System includes the complete chassis, and video monitor as pictured below.



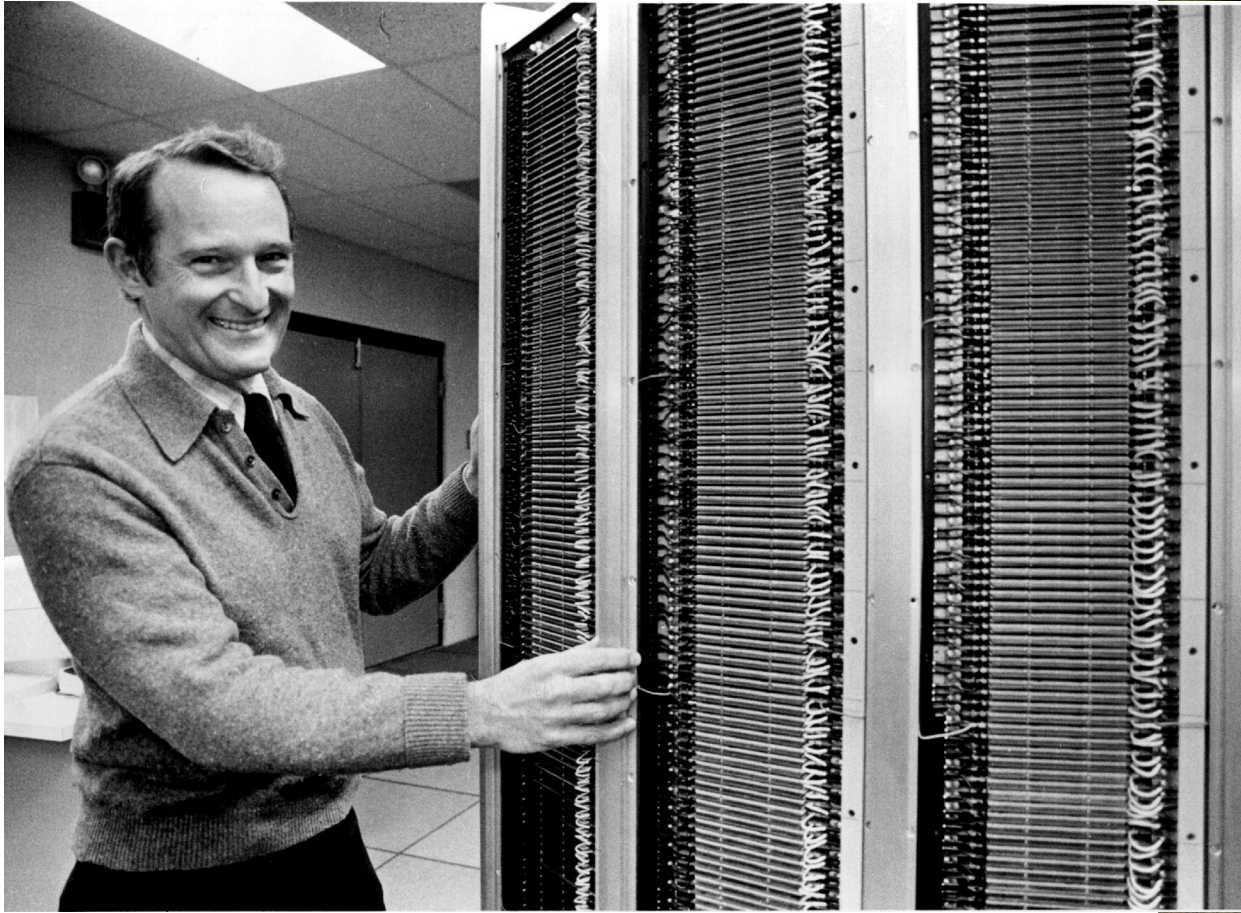
Mike Wise





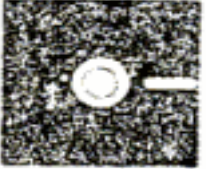
Bill Mensh (Estados Unidos, 1945)

8.35 Supercomputadores CRAY – 1976



Seymour Cray (Estados Unidos, 1925-1996)

8.36 Primeiro Sistema Operacional para Microprocessador: CP/M – 1976



CP/M™
LOW-COST
MICROCOMPUTER
SOFTWARE

CP/M™ OPERATING SYSTEM:

- Editor, Assembler, Debugger and Utilities.
- For 8080, Z80, or Intel MDS.
- For IBM-compatible floppy discs.
- **\$100**-Diskette and Documentation.
- **\$25**-Documentation (Set of 6 manuals) only.

MAC™ MACRO ASSEMBLER:

- Compatible with new Intel macro standard.
- Complete guide to macro applications.
- **\$90**-Diskette and Manual.

SID™ SYMBOLIC DEBUGGER

- Symbolic memory reference.
- Built-in assembler/disassembler.
- **\$75**-Diskette and Manual.

TEX™ TEXT FORMATTER

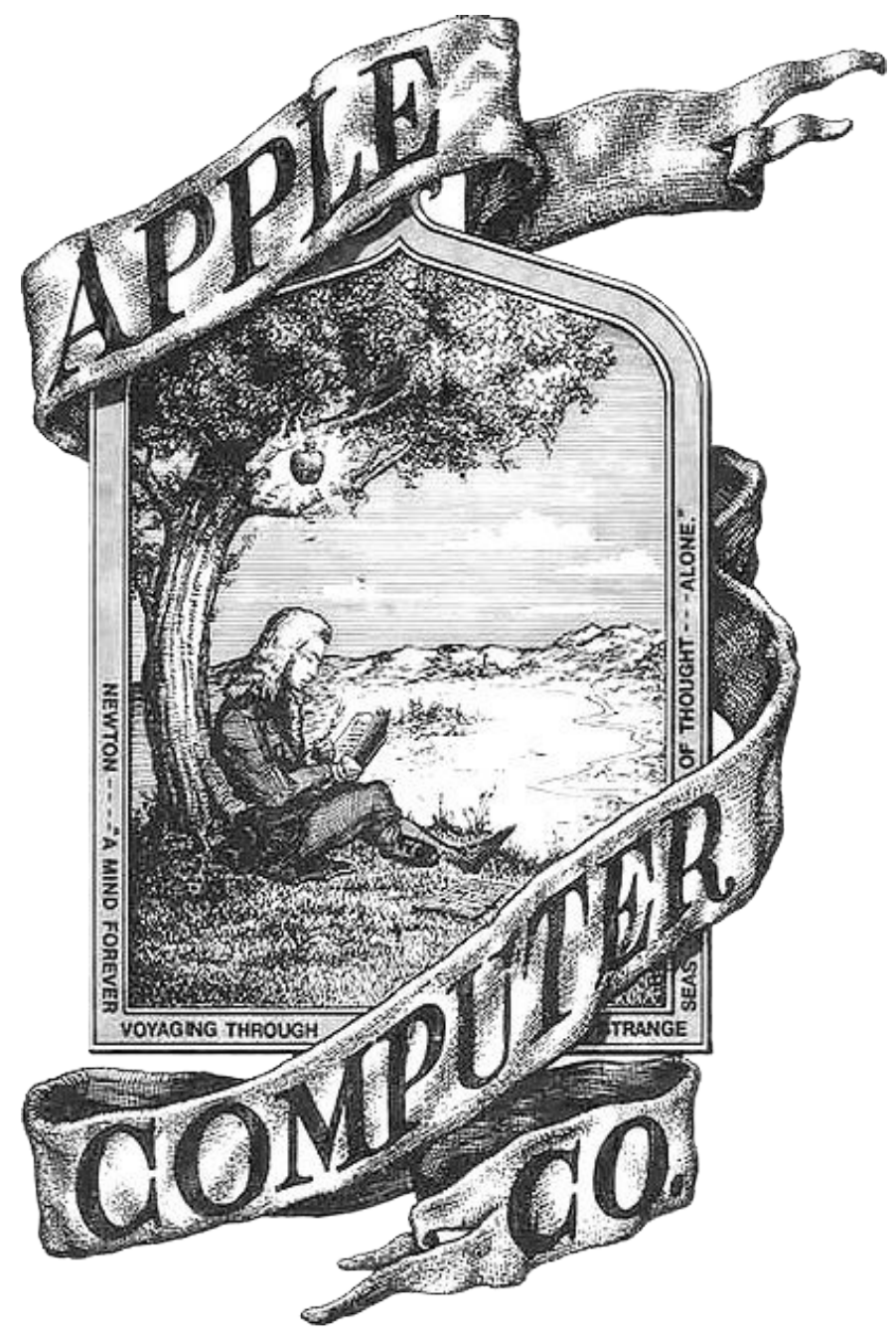
- Powerful text formatting capabilities.
- Text prepared using CP/M Editor.
- **\$75** Diskette and Manual.

 **DIGITAL RESEARCH**
P.O. Box 579 • Pacific Grove, CA 93950
(408) 649-3896



Gary Kildall (Estados Unidos, 1942-1994)

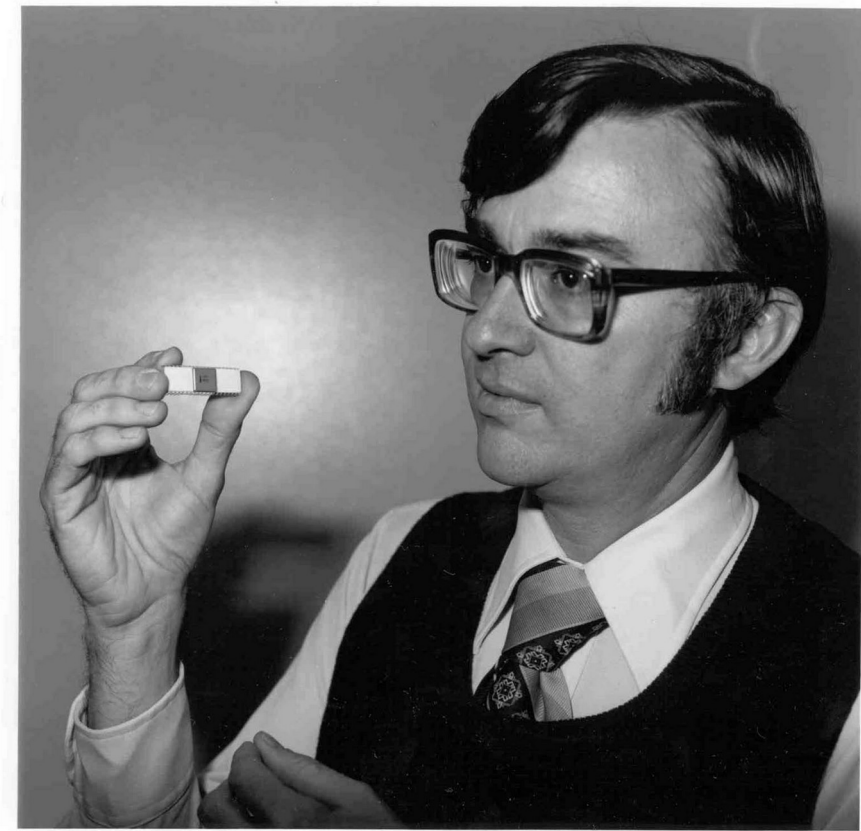
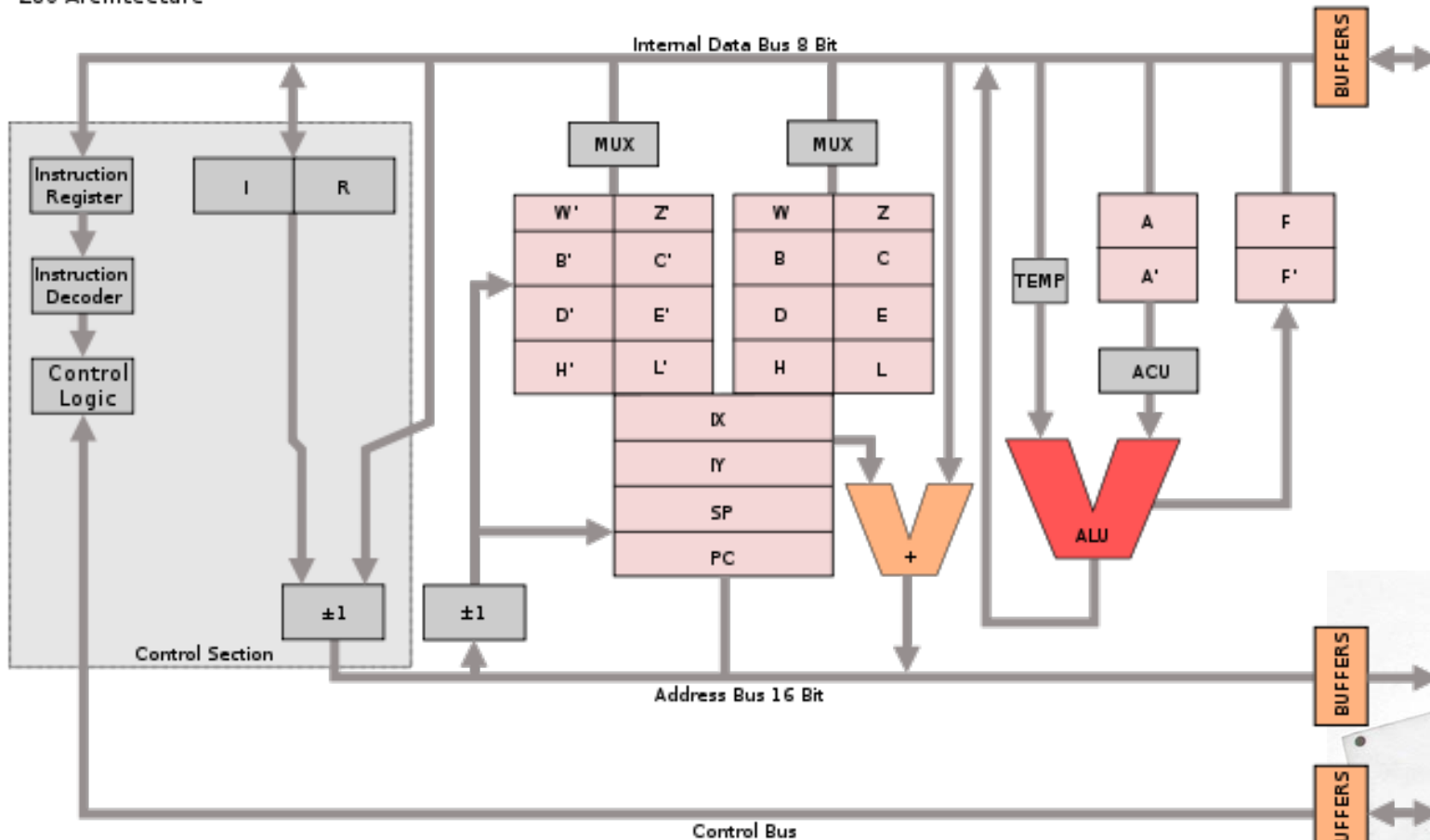
8.37 Apple 1 – 1976



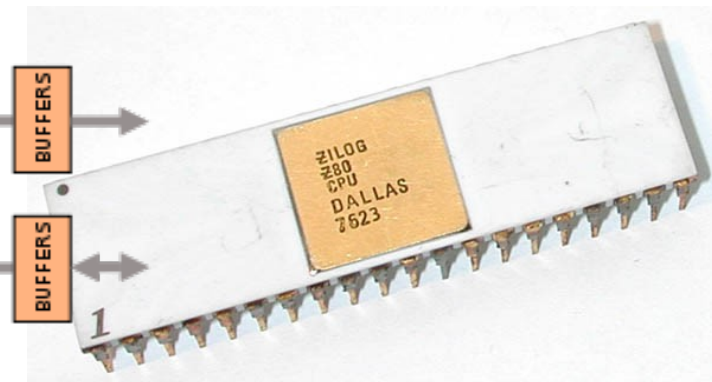
Steve Jobs (Estado Unidos, 1955-2011) e Steve Wozniak (Estados Unidos, 1950)

8.38 Zilog Z80 – 1976

Z80 Architecture



Federico Faggin (Itália, 1941)



8.39 Sol 20 – 1976

Processor Technology Corporation

Gary Ingram e Bob Marsh

```
CHES 1.0 (MOD 0)
DO YOU WANT BLACK (B) OR WHITE (W) ?

      PLAYER  COMPUTER
01: D2D4    D7D5
02: B1C3    B8C6
03: C1F4    G8F6
04:
```

```
--- 8080 CHES ---
ProcessorTechnology
-A-B-C-D-E-F-G-H-
8: R B Q K B R 18
7: P P P P P P P 17
6: N N 16
5: P 15
4: P B 14
3: N 13
2: P P P P P P 12
1: R Q K B N R 11
-A-B-C-D-E-F-G-H-
```

C1F4.



The Sol Computer



The Sol Computer (Inside)

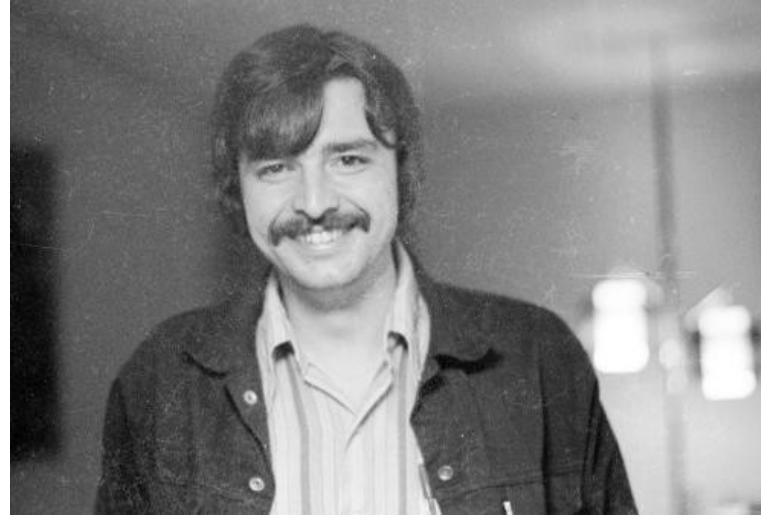
8.40 CompuColor 8001, o primeiro computador a cores – 1976



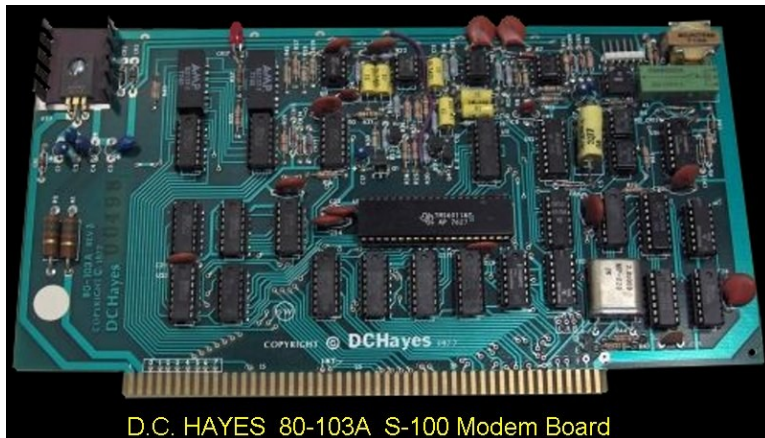
8.41 Modem 80-103A – 1977



Dale Heatherington



Dennis Hayes (Estados Unidos, 1950)



D.C. HAYES 80-103A S-100 Modem Board



8.42 DEC VAX 11 – 1977



8.43 Apple II - 1977



Jobs e Wozniak



8.44 Tandy TRS80 – 1977



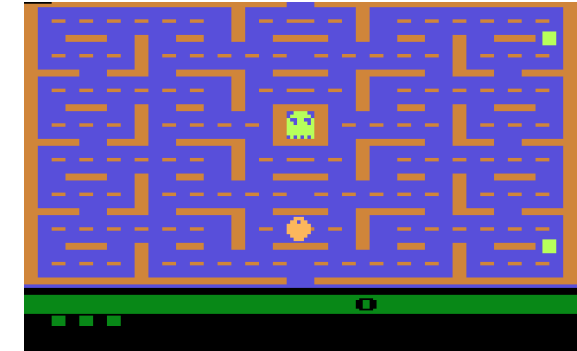
Charles D. Tandy (Estados Unidos, 1918-1978)

8.45 Commodore PET - 1977

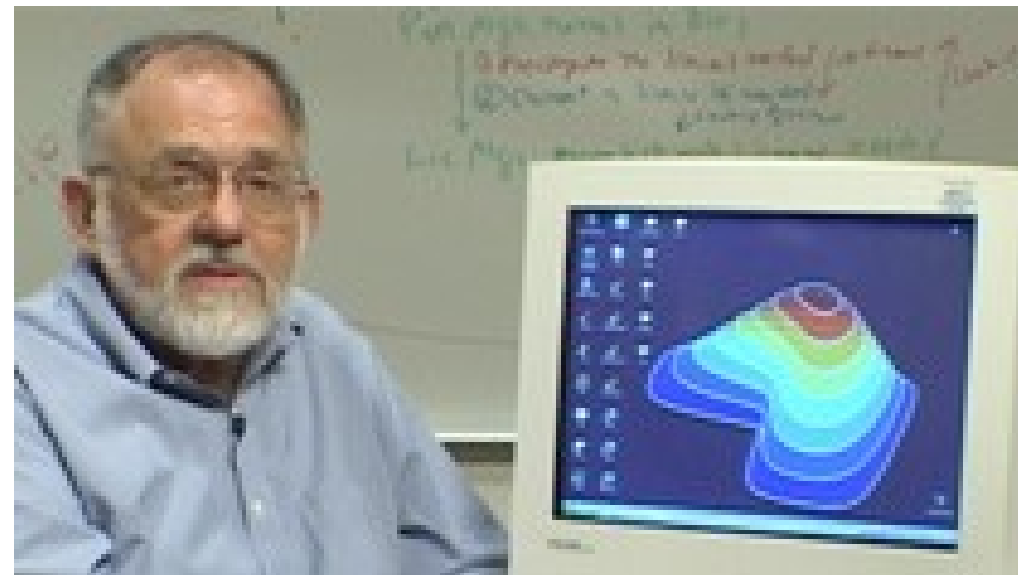
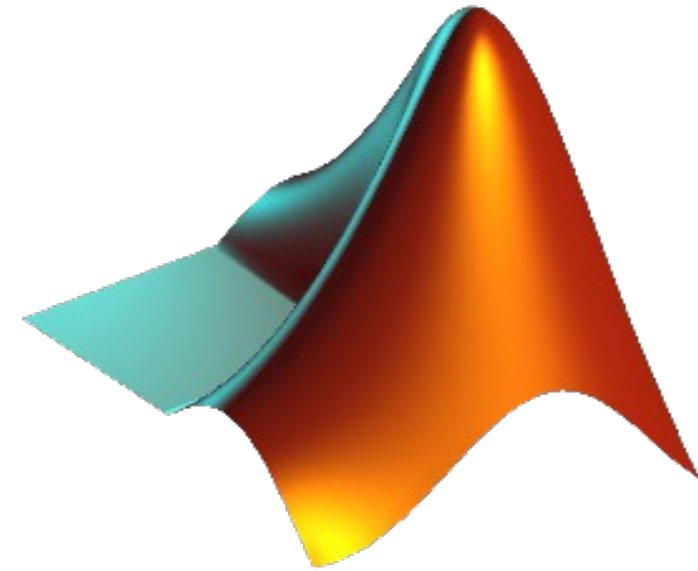
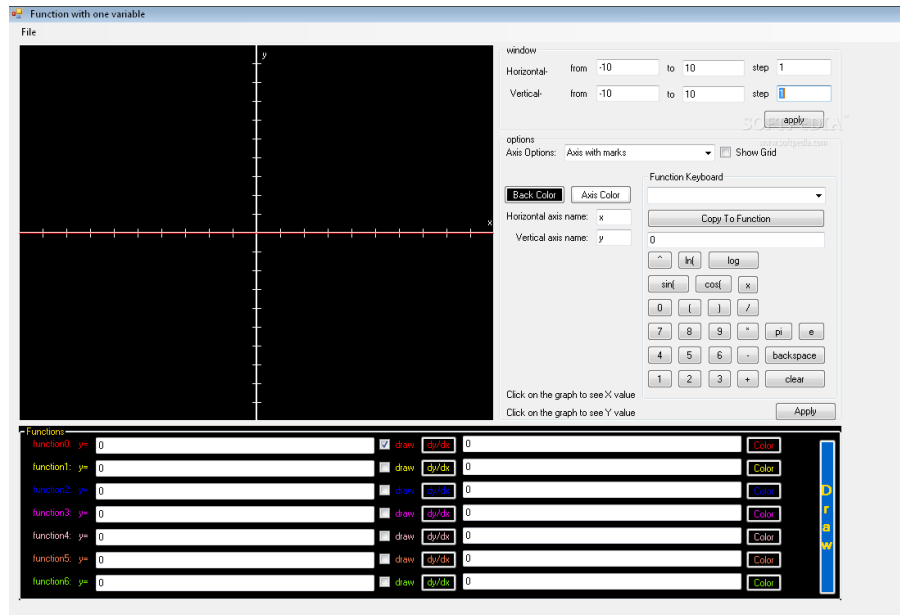


Jack Tramiel (Polônia, 1928-2012)

8.46 Console de Videogame Atari 2600 – 1977

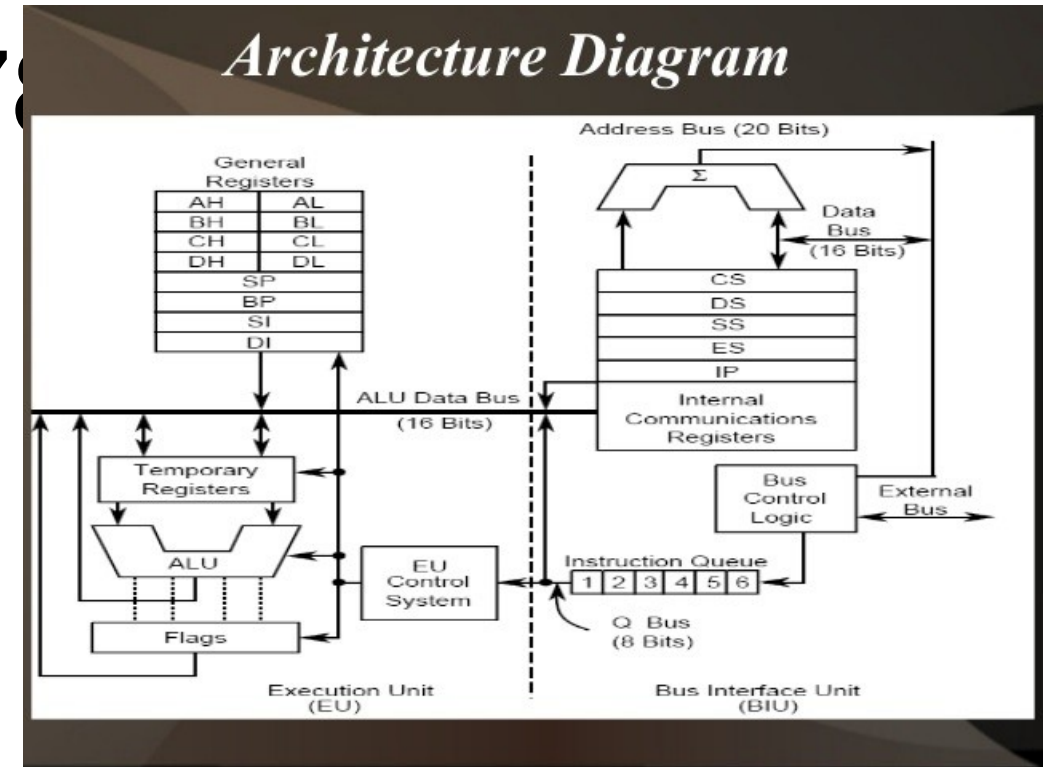
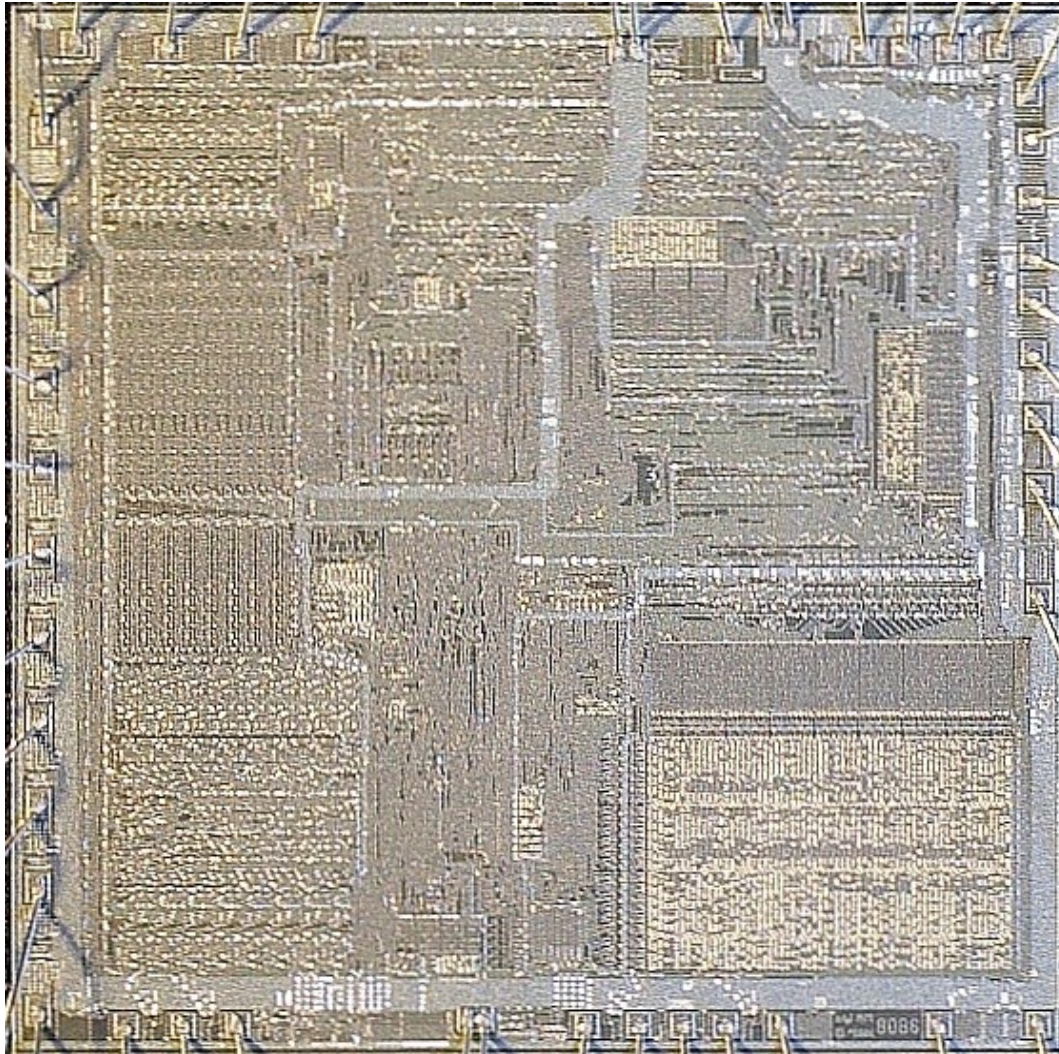


8.47 Matlab – 1978



Clever Moler (Estados Unidos, 1939)

8.48 Intel 8086 – 1978



8.49 Xerox NoteTaker – 1978

Xerox NoteTaker

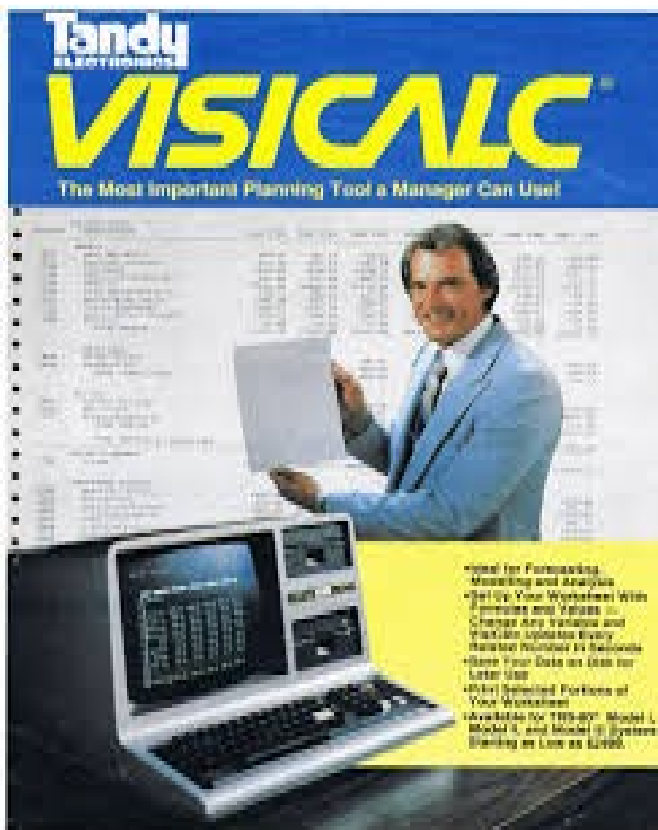


The first laptop. But it couldn't really fit on your lap.



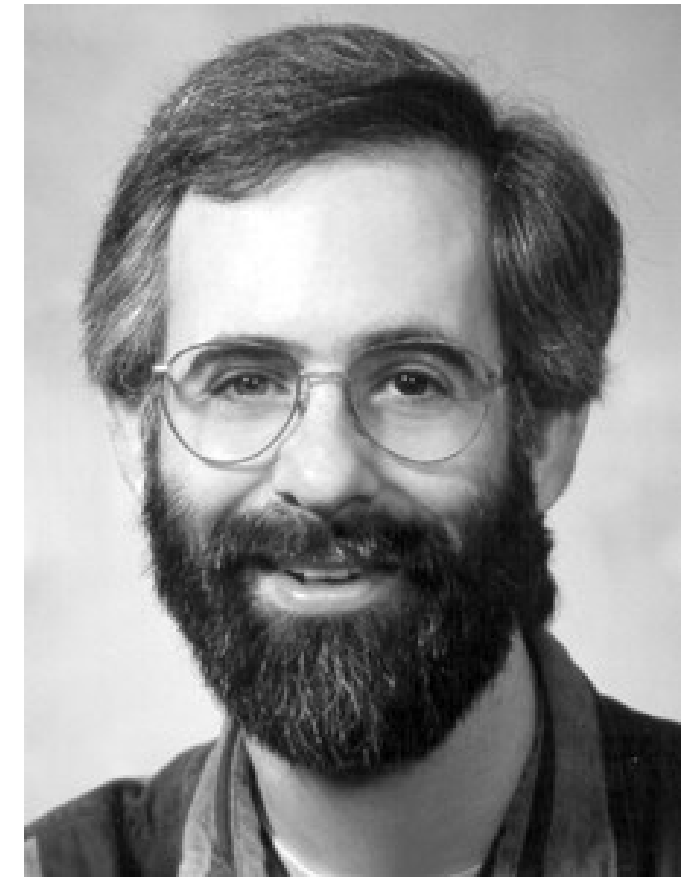
Douglas Fairbairn

8.50 Planilha Eletrônica VisiCalc – 1978



C11 (L) TOTAL C1 25

	A	B	C	D
1	ITEM	NO.	UNIT	COST
2	MUCK	43	12.95	556.85
3	BUZZ	15	6.75	101.25
4	TOE	250	49.95	12487.50
5	EYE	2	4.95	9.90
6				
7			SUBTOTAL	13155.50
8			9.75% TAX	1282.66
9			TOTAL	14438.16



Daniel S. Bricklin (Estados Unidos, 1951)

8.51 SBC - Sociedade Brasileira de Computação – 1978



CSBC 2016

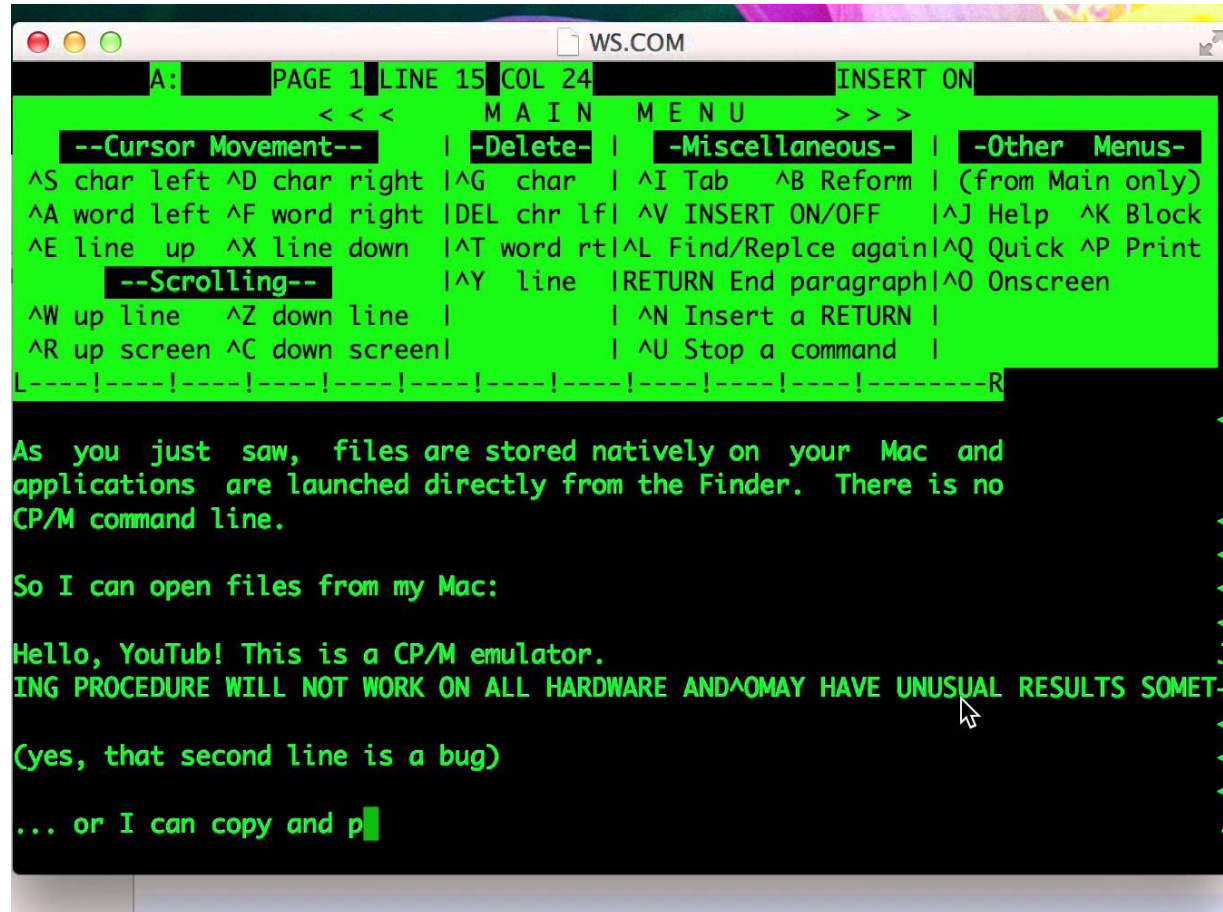
XXXVI Congresso da Sociedade Brasileira de Computação

#ComputaçãoInterdisciplinaridade

04 A 07 DE JULHO | PUCRS | PORTO ALEGRE/RS



8.52 Editor de Texto WordStar – 1979



```
A: PAGE 1 LINE 15 COL 24 INSERT ON
< < < MAIN MENU > > >
--Cursor Movement-- | -Delete- | -Miscellaneous- | -Other Menus-
^S char left ^D char right | ^G char | ^I Tab ^B Reform | (from Main only)
^A word left ^F word right | ^DEL chr lfl | ^V INSERT ON/OFF | ^J Help ^K Block
^E line up ^X line down | ^T word rt | ^L Find/Replce again | ^Q Quick ^P Print
--Scrolling-- | ^Y line | ^RETURN End paragraph | ^O Onscreen
^W up line ^Z down line | | ^N Insert a RETURN |
^R up screen ^C down screen | | ^U Stop a command |
L-----!-----!-----!-----!-----!-----!-----!-----!-----!-----R

As you just saw, files are stored natively on your Mac and
applications are launched directly from the Finder. There is no
CP/M command line.

So I can open files from my Mac:

Hello, YouTub! This is a CP/M emulator.
ING PROCEDURE WILL NOT WORK ON ALL HARDWARE AND^OMAY HAVE UNUSUAL RESULTS SOMET+
(yes, that second line is a bug)

... or I can copy and p
```

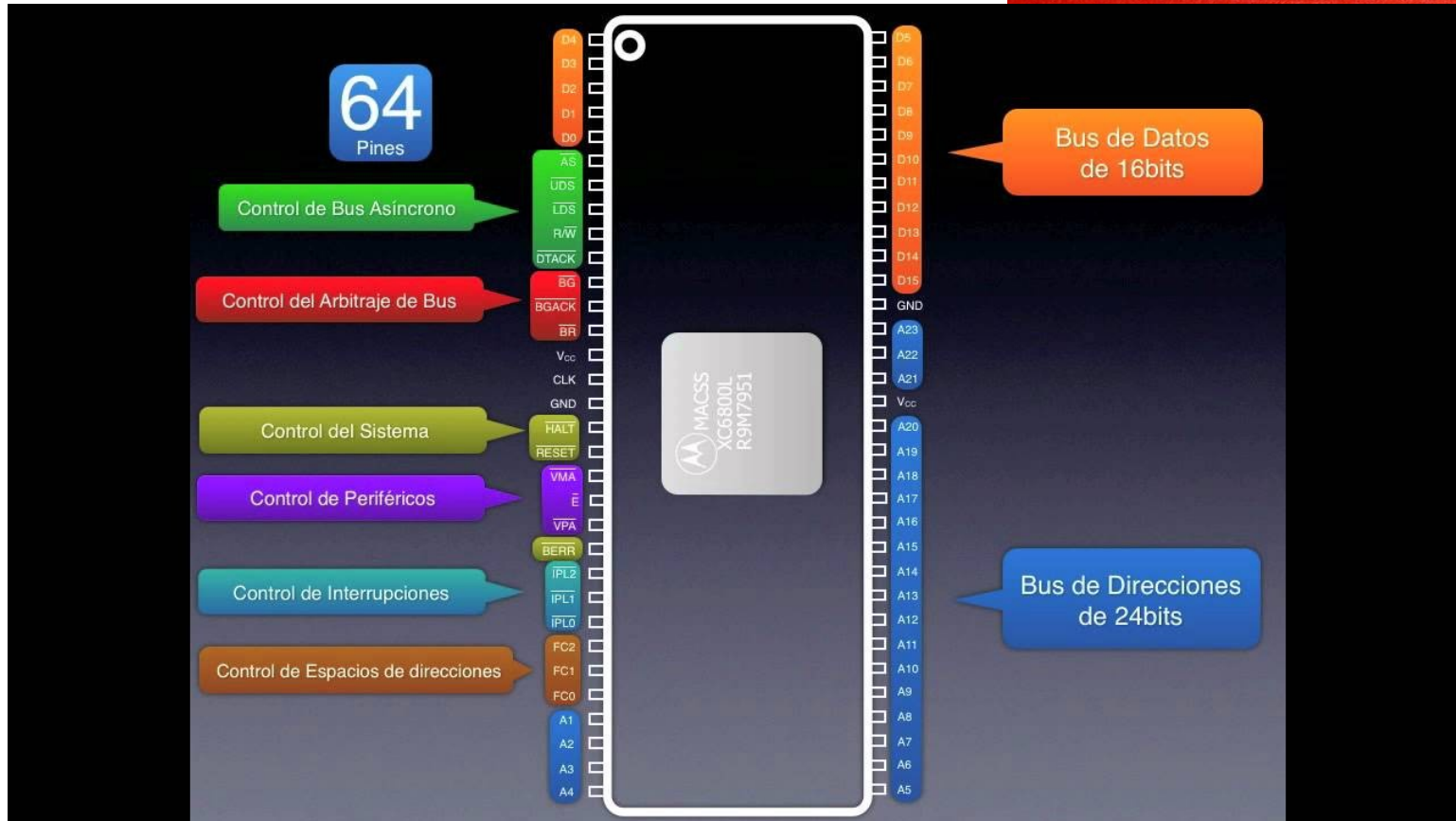
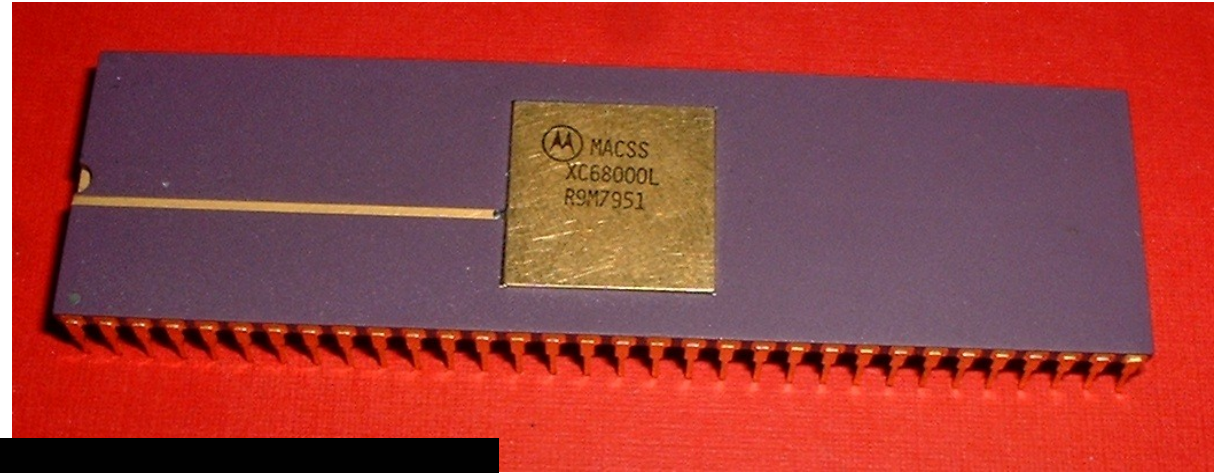


Rob Barnaby



Seymour I. Rubinstein (Estados Unidos, 1934)

8.53 Motorola 68000 – 1979



8.54 SQL – 1979



```
SELECT EmployeeID, FirstName, LastName, HireDate, Ci  
FROM Employees  
WHERE HireDate BETWEEN '1-june-2012' AND '15-decem
```

```
SELECT EmployeeID, FirstName, LastName, HireDate, City  
WHERE City IN ('Seattle', 'Tacoma', 'Redmond')
```

```
SELECT EmployeeID, FirstName, LastName, HireDate, Ci  
FROM Employees  
WHERE HireDate NOT BETWEEN '1-june-2012' AND '15-de
```

Donald D. Chamberlin (Estados Unidos, 1944) e Raymond F. Boyce (Estados Unidos, 1947-1974)

Até aqui...

- Vimos que os anos 1970 testemunharam o surgimento de computadores pessoais realmente úteis e com aplicativos como editores de texto e planilhas eletrônicas que os tornavam inclusive uma opção bem mais barata para uso em empresas.
- Isso foi possível a partir da concepção de microprocessadores como o 4004, 8008 e 8080 da Intel. Estes ainda eram relativamente caros, mas com o lançamento do MOS 6502 a meros 25 dólares, microcomputadores suficientemente baratos e mesmo consoles de videogames tornaram-se possíveis.
- Porém, raramente estes computadores se ligavam em rede. Isso foi algo que só começou a se tornar usual na década seguinte.
- A ARPANET e outras redes ao redor do mundo seguiam crescendo, interligando principalmente órgãos governamentais e universidades.
- A invenção do TCP/IP nos anos 1970 permitiu que essas redes se interligassem, dando início ao fenômeno da Internet. Mas o usuário doméstico ainda não tinha acesso a ela. A invenção do modem para microcomputador, porém, deu início a essa possibilidade.
- Em termos de interfaces, as ideias de Vannevar Bush e Ivan Sutherland continuavam atraentes e com isso começaram a engatinhar projetos de computadores com interfaces gráficas, mas o custo e a baixa tecnologia da época ainda não permitiam que isso fosse universalizado.
- Já havia computadores com capacidade gráfica e cores, inclusive, mas essas capacidades usualmente só eram utilizadas em jogos, a maioria dos quais muito simples. Um ambiente se destacava, Smalltalk do Xerox Parc, uma linguagem de programação orientada a objetos totalmente baseada em ambiente gráfico. Mas seu uso era ainda muito restrito.
- No Brasil, os anos 1970 viram o surgimento do Pato Feio e da COBRA, a primeira indústria com tecnologia nacional. Surge também a SBC, Sociedade Brasileira de Computação, a partir da consolidação de eventos científicos que já se realizavam no país.